Environmental influences on Child Health Outcomes (ECHO) Cohort Data and Biospecimen Collection Protocol

Version Number: 3.2
Date: 06 November 2024
Funding Sponsor: National Institutes of Health (NIH)

ECHO Program Director	Matthew W. Gillman, MD, SM Director, Environmental influences on Child Health Outcomes (ECHO) Office of the Director, National Institutes of Health Phone: (301) 402-3388 Email: matthew.gillman@nih.gov

Executive Summary

The Environmental influences on Child Health Outcomes (ECHO) Program seeks to address the impact of a broad range of early environmental influences on child development and health. The ECHO Cohort Consortium brings together data and biospecimens from Cohort Study Sites across the United States on various environmental exposures, e.g., physical, chemical, social, behavioral, and biological, to support numerous longitudinal analyses. These ECHO Cohort Study Sites will share standardized data and biospecimen collection methods and focus on five key outcome domains: pre-, peri-, and postnatal outcomes; neurodevelopment; upper and lower airways; obesity and obesity-related conditions; and positive health.

The primary objective of the ECHO Cohort Data and Biospecimen Collection Protocol (ECHO Cohort Protocol) is to facilitate the continual updating of the ECHO Cohort Data Platform consisting of core data elements from all ECHO Cohort Study Sites. ECHO Cycle 2 is a continuation of ECHO Cycle 1, which initiated the ECHO Cohort Data Platform. This protocol serves to standardize new data collection across the ECHO Cohort Program and to specify what data elements and biospecimens ECHO Cohort Study Sites should collect and share across the participant life course – from the prenatal period through adolescence – to create the ECHO Cohort.

ECHO Cohort Investigators and the wider scientific community can use the ECHO Cohort data to address innovative, high-impact, solution-oriented research questions that require multiple Cohort Study Sites and investigate the relationships of a broad range of early environmental influences to child health outcomes.

Protocol Version and Amendment Tracking

Version No.	NIH Approval Date
1.0	18 April 2018
1.1	03 August 2018
1.2	11 December 2018
1.3	09 April 2020
2.0	24 February 2021
2.1	25 June 2021
3.0	24 May 2023
	25 September 2023
3.1	21 August 2024
3.2 (see Section 10.2 for a summary of changes from Version 3.1 to Version 3.2)	06 November 2024

Protocol Synopsis

Protocol Version	**Version 3.2**
Protocol Title	**Environmental influences on Child Health Outcomes (ECHO) Cohort Data and Biospecimen Collection Protocol**
Objective	The ECHO Program seeks to address the impact of a broad array of early environmental influences on child development and health. The ECHO Cohort Data and Biospecimen Collection Protocol will facilitate the generation of the ECHO Cohort Data Platform consisting of core data elements from all Cohort Study Sites. This protocol specifies the data and biospecimens that ECHO Cohort Study Sites will collect and share to create the ECHO Cohort Data Platform. This protocol serves to standardize data and biospecimen collection across the ECHO Cohort Program and will evolve over the course of the ECHO Program.
Study Design	The ECHO Cohort is a longitudinal prospective study consisting of Cohort Study sites across the United States, including Puerto Rico, which collect data and biospecimens throughout a child's life course, from (pre-) pregnancy through age 20 years. The ECHO Cohort Data and Biospecimen Collection Protocol supports the standardized data and biospecimen collection methods through the creation of the ECHO Cohort Data Platform.

Inclusion Criteria and Research Participants	This protocol applies to all participants within Cohort Study Sites of ECHO Cohort Awards (ECHO Cohort Study Sites) who meet at least one of these five criteria for an ECHO Cohort Data and Biospecimen Collection Protocol participant: 1. Child enrolled in an ECHO Cohort Study Site who meet the criteria: a) children born to pregnant enrollees in ECHO Cycle 2, or b) children who were enrolled for new data collection in ECHO Cycle 1. 2. Pregnant person enrolled at less than 20 weeks of pregnancy in an ECHO Cohort Study Site* 3. Primary caregiver, preferably the biological mother, of the child (or children) anticipated to enroll by an ECHO Cohort Study Site. An ECHO Cohort Study Site that did not previously enroll the child's primary caregiver will be required to seek their permission for participation under the ECHO Cohort Data and Biospecimen Collection Protocol. 4. Biological mother of a child (or children) anticipated to enroll in an ECHO Cohort Study Site, if not already enrolled under inclusion criterion #3 above, for the collection of "sample appropriate for deoxyribonucleic acid (DNA)." 5. Current partner of an enrolled participant who completed an ECHO pregnancy within the prior eleven months, thirty days. ECHO will follow children and pregnant persons for ECHO outcomes of interest. Primary caregivers (and biological mothers, as applicable) and current partners will provide valuable biospecimens and exposure/covariate/outcome-related data through self-report and proxy-

	report measures that will help ECHO understand how early environmental exposures influence child health outcomes. ECHO Cohort Study Sites will seek consent (and permission/assent as appropriate) for the participation of all possible participants noted above and the collection of their data. Cohort Study Sites will also collect data from consented participants about other individuals who interact with ECHO Cohort Data and Biospecimen Collection Protocol participants. These individuals may include biological parents and siblings, non-biological parents and siblings, and other caregivers of the child, as well as partners, biological parents and siblings, and other children of the pregnant person.
Exclusion Criteria	A surrogate pregnancy is excluded from enrollment.
Duration of Study Participation	The funding period for the ECHO Cohort Cycle 2 is seven years, starting in September 2023 (FY23 funds). Under this cycle, ECHO Study Sites will collect data and biospecimens through August 2030.
Number of Cohorts	In Cycle 1, September 2016 to August 2023, the ECHO Program consisted of 31 Cohort Awardees that comprised 69 cohorts and their associated participating ECHO cohort sites (approximately 180 sites). In Cycle 2, from September 2023, the ECHO Cohort will consist of ~50 Cohort Study Site Awards with follow-up of existing ECHO child participants and recruitment of ~ 20,000 pregnant participants, yielding over 50,000 children throughout the ECHO Program.

Number of Cohort Study Sites	A Cohort Study Site is “an institution at which participant recruitment or follow-up takes place at one or multiple locations. E.g., a Cohort Study Site could be a medical system with participant follow-up taking place at its clinics. In Cycle 2, from September 2023, the ECHO Cohort will consist of ~50 Cohort Study Site Awards.
Number of Participants	ECHO expects the number of participants to be approximately 60,000 children and almost as many primary caregivers over the course of the ECHO Program.

Contact Information

Cohort Study Sites may direct questions regarding the procedures, processes, or responsibilities described in the ECHO Cohort Data and Biospecimen Collection Protocol to their Program Officers.

Table of Contents

Executive Summary **2**
Protocol Version and Amendment Tracking **3**
Protocol Synopsis **4**
Contact Information **8**
Table of Contents **9**
List of Tables **12**
List of Figures **12**
Abbreviations **13**
ECHO Cohort Data and Biospecimen Collection Protocol Glossary **14**
1. Background and Rationale **21**
2. Objective and Development of the ECHO Cohort Data and Biospecimen Collection Protocol **25**
2.1 Primary Objective of the ECHO Cohort Data and Biospecimen Collection Protocol 25
2.2 Development of the ECHO Cohort Data and Biospecimen Collection Protocol 26
2.3 Development of the ECHO Streamlined Protocol 27
3. ECHO Program Participant Population **30**
3.1 Inclusion Criteria 30
3.2 Exclusion Criteria 31
3.3 Duration of Participation 31
3.4 Number of Participants 31
3.5 Participant Withdrawal 31
3.6 ECHO Cohort Data and Biospecimen Collection Protocol Design: Data Elements and Biospecimens 31
4. ECHO Cohort Data and Biospecimen Collection Protocol Procedures **32**
4.1 Frequency of Data Collection 34
4.2 Core Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course 34

4.3 Specialized Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course 35

5. Data Management 167
5.1 Data Submission to the ECHO Data Analysis Center (DAC) 167
5.2 ECHOPortal and Data Security 167
5.3 Data Quality Assurance 168
5.4 Data Access 169
5.5 Biospecimen Tracking 170
5.6 Record Retention 170

6. Analysis Proposals 170

7. Study Responsibilities 170
7.1 ECHO Coordinating Center Responsibilities 170
7.2 ECHO Data Analysis Center Responsibilities 171
7.3 ECHO Measurement Core Responsibilities 171
7.4 ECHO Cohort Study Site Investigator Responsibilities 172
7.5 ECHO Laboratory Core Responsibilities 172

8. Ethical Considerations and Human Subjects Protection 172
8.1 Institutional Review Board/Ethics Committee Review 172
8.2 Informed Consent 173
8.3 Confidentiality of Participants 174
8.4 Authorization for Use and Disclosure of Protected Health Information (HIPAA) 174
8.5 Risks to ECHO Cohort Data Collection Protocol Participants 174
8.6 Reimbursement to ECHO Cohort Data Collection Protocol Participants 175

9. Regulatory Reporting 175
9.1 Unanticipated Problems Involving Risks to Subjects or Others 175
9.2 Breach Notifications 175
9.3 Reports of Non-compliance 176

10. Appendices 177
10.1 ECHO Policies 177

10.2 ECHO Cohort Data Collection Protocol Version 3.1 to Version 3.2: Summary of Changes 178

List of Tables

Table 1 **Data Elements for First Visit** 37
Table 2 **Data Elements for Year 2024** 38
Table 3 **Data Elements for Year 2025** 52
Table 4 **Data Elements for Year 2026** 65
Table 5 **Data Elements for Year 2027** 78
Table 6 **Data Elements for Year 2028** 93
Table 7 **Data Elements for Year 2029** 106
Table 8 **Data Elements for Year 2030** 119
Table 9 **Specialized Data Elements for Year 2025** 134
Table 10 **Specialized Data Elements for Year 2026** 140
Table 11 **Specialized Data Elements for Year 2027** 145
Table 12 **Specialized Data Elements for Year 2028** 150
Table 13 **Specialized Data Elements for Year 2029** 155
Table 14 **Specialized Data Elements for Year 2030** 162

List of Figures

Figure 1 **ECHO Program Organizational Structure** 23

Abbreviations

CAT	Computer Adaptive Test
CC	Coordinating Center
DAC	Data Analysis Center
ECHO	Environmental influences on Child Health Outcomes
FISMA	Federal Information Security Management Act
HIPAA	Health Insurance Portability and Accountability Act
IRB	Institutional Review Board
MOP	Manual of Operating Procedures
NIH	National Institutes of Health
PHI	Protected Health Information
PIE	ECHO Cohort Data and Biospecimen Collection Protocol Implementation and Evaluation Committee
PRO	Person-Reported Outcome
QA	Quality Assurance
QC	Quality Control
sIRB	Single Institutional Review Board
UP	Unanticipated Problem

ECHO Cohort Data and Biospecimen Collection Protocol Glossary

The following table contains a glossary of specialized terminology used within the ECHO Cohort Data and Biospecimen Collection Protocol.

Term	Definition
Biospecimen	Materials taken from the human body, e.g., blood, urine, hair, stool.
Biospecimens Working Group	The ECHO working group charged with developing policies and procedures for biospecimen collection, processing, storage, and utilization, and with exploring novel related methodologies.
Chemical Exposures Working Group	The ECHO working group charged with recommending analyses of chemical exposures.
Core Data Element	Standardized key terms or concepts in this protocol to enhance data quality so that researchers can use data more effectively across analyses and over time. Core data elements promote data standardization and comparability across research studies. ECHO Program expects Cohort Study Sites to collect all core data elements on all participants appropriate to their age, as specified in this protocol. Core elements in the ECHO Cohort Data and Biospecimens Collection Protocol include all of the elements in Section 4.2 below.
Coordinating Center (CC)	The ECHO component that provides comprehensive operational leadership and an organizational infrastructure to manage and coordinate all ECHO Cohort activities.

Term	Definition
Data Analysis Center (DAC)	The ECHO component that leads, standardizes, and integrates ECHO Cohort Protocol data capture, management, and storage through a central data system; provides analytic support and expertise to analysis proposals approved by the ECHO Cohort consortium; and enriches research infrastructure and data science to facilitate broader sharing of ECHO Cohort data and resources with the scientific community.
Data Collection Form	Data collection forms provide specifications for how ECHO Cohort Study Sites will collect related sets of variables (“data elements”) on the ECHO Data Platform at the form, section, and item level. Data collection forms have associated data dictionaries that describe attributes of each data element (e.g., the element's internal name, data type, caption presented to users, and basic validation information such as range checks), skip logic, and expressions for calculated elements.
ECHO Manual of Operating Procedures (MOP)	A handbook that supplements the ECHO Cohort Data and Biospecimen Collection Protocol to guide study conduct and operations. ECHO Cohort Study Sites will use the MOP as a reference document for procedures related to the ECHO Cohort Data and Biospecimen Collection Protocol.
ECHO Cohort Consortium	All Cohort Study Sites plus all Cores and Centers
ECHO Cohort Data Platform	A data platform consisting of core data elements from all Cohort Study Sites.
ECHO Cohort Analysis	An analysis that uses the ECHO Cohort.
ECHO Cohort Data and Biospecimen Collection Protocol	The protocol that specifies what data and biospecimens Cohort Study Sites should collect and share across the life course from the prenatal period through adolescence. This protocol facilitates the continuation of the ECHO Cohort Data Platform.

Term	Definition
ECHO Cohort Data and Biospecimen Collection Protocol Development Working Group	The ECHO working group responsible for the development of the initial version(s) of the ECHO Cohort Data and Biospecimen Collection Protocol.
ECHO Cohort Data and Biospecimen Collection Protocol Implementation and Evaluation Committee (PIE)	The ECHO committee responsible for improving and evaluating the implementation feasibility of the ECHO Cohort Data Collection and Biospecimen Protocol.
ECHO Investigator	ECHO Award Principal Investigator or Co-Investigator
ECHO Cohort Operations Committee	This ECHO committee will enable the achievement of the ECHO Cohort Steering Committee's scientific priorities and the operational objectives that the NIH ECHO Program Office sets. The committee will also facilitate the function of a central dashboard that displays real-time consortium-wide progress toward the annual ECHO Cohort consortium's scientific and operational objectives.
ECHO Cohort Steering Committee	Represents the interests of all ECHO components, provides overarching strategic direction for the ECHO Program within the vision and rules stipulated by NIH, and guides the scientific work of the ECHO Program.
ECHO Cohort Study Site	An institution at which participant recruitment and follow-up takes place at one or multiple locations.
ECHO Cohort Study Site Pods	The ECHO Cohort governance structure will include multiple small groups of ECHO Cohort Study Site PDs/PIs of approximately equal size ("Pods") to enhance bi-directional communication between them and the ECHO Cohort Steering Committee and Operations Committee.
ECHO External Scientific Board	The External Scientific Board will review the ECHO Cohort consortium's progress and provide feedback to the NIH ECHO Director, NIH ECHO Program staff, and the appropriate NIH Advisory Council on the ECHO Cohort consortium's programmatic strategies.
ECHO Program Office	The NIH office that provides oversight, and scientific and programmatic coordination.

Term	Definition
ECHOPortal	A gateway for data capture and analysis that provides a collaborative work environment for use of ECHO Cohort data.
Exposure	A determinant, risk or resilience factor, predictor, or independent variable associated with child health. Exposures can be from a broad range, including physical, chemical, social, behavioral, or biological. ECHO defines the early exposure period from preconception to age 5 years.
Laboratory Core	The component of ECHO that provides a focused infrastructure for comprehensive environmental chemical exposure analysis of biological samples: 1. Rigorous assessment of a range of environmental exposures, xenobiotics, physiological measurements, metabolomics, nutrients, inflammatory biomarkers, and other biological indicators of environmental exposure. 2. Statistical tools and data science approaches to manage and analyze newly generated datasets pertaining to chemical exposure analysis of biological samples in a cohesive and integrated manner.
Neurodevelopment Outcome Domain	The outcome domain focused on neurodevelopmental outcomes (e.g., attention, cognition, emotion, social/language/behavioral development).
Obesity Outcome Domain	The outcome domain focused on obesity and obesity-related outcomes (e.g., metabolic risk factors).
Outcome Domains	ECHO studies will focus on five key child health outcome domains that have a high public health impact: • Pre-, peri-, postnatal outcomes • Neurodevelopment • Upper and lower airways • Obesity and obesity-related conditions • Positive health

Term	Definition
Outcome Working Groups	Groups of experts representing the five ECHO outcome domains, charged with developing outcome-focused research questions and identifying data elements and appropriate measurements to inform the development of the ECHO Cohort Data and Biospecimen Collection Protocol.

Term	Definition
Participants	The ECHO Cohort Data and Biospecimen Collection Protocol defines eligible research participants as: 1. Child enrolled in an ECHO Cohort Study Site* who meet the criteria: a) children born to pregnant enrollees in ECHO Cycle 2, or b) children who were enrolled for new data collection in ECHO Cycle 1. 2. Pregnant person enrolled at less than 20 weeks of pregnancy in an ECHO Cohort Study Site.* 3. Primary caregivers, preferably the biological mother, where possible, of children anticipated to enroll by ECHO Cohort Study Sites. An ECHO Cohort Study Site cohort that did not previously enroll the child's primary caregiver must seek their permission for participation under the ECHO Cohort Data Collection Protocol. 4. Biological mother of a child (or children) anticipated to enroll by an ECHO Cohort Study Site, if not already enrolled under inclusion criterion #3 above, for the collection of "sample appropriate for DNA." 5. Current partner of an enrolled participant who completed an ECHO pregnancy within the prior eleven months, thirty days. * ECHO will follow children and pregnant persons for ECHO outcomes of interest. Primary caregivers (and biological mothers, as applicable) and current partners will provide valuable biospecimen and exposure/covariate/outcome-related data through self-report and proxy-report measures that will help ECHO understand how early environmental exposures influence child health outcomes. * ECHO Cohort Study Sites will seek consent (and permission/assent as appropriate) for their participation and collection of data. Cohort Study Sites will also collect data from consented participants about other individuals who interact with ECHO Cohort Data and Biospecimen Collection Protocol participants. These individuals may include biological parents and siblings, nonbiological parents and siblings, and other caregivers of the child; and partners, biological parents and siblings, and other children of the pregnant person.

Term	Definition
Measurement Core	The component of ECHO that provides measurement expertise in the development, selection, and validation of self and proxy-reported and performance-related measures as assessed from children and adults in the ECHO Cohort Study Sites.
Positive Health Outcome Domain	The outcome domain focused primarily on psychological well-being, as measured by global health, life satisfaction, and meaning and purpose, reflecting an individual's capacity to adapt, satisfy needs, and fulfill goals.
Pre-, Peri-, Postnatal Outcome Domain	The outcome domain focused on pre-, peri-, and postnatal outcomes (e.g., birth size, prematurity, neonatal/infant mortality).
Primary Caregiver	A primary caregiver is an individual, such as a family member (biological or non-biological) or guardian, who is responsible for the care of the child.
Protocol Task Force (PTF)	ECHO group tasked with reviewing and modifying the existing protocol to create a single protocol without excessive options that balances burden with scientific value for use in NIH-funded longitudinal studies.
Specialized Data Element	Specialized Data Elements include more in-depth or frequent elements than the Core elements. They are organized into three Exposure Areas (Chemical/Physical, Lifestyle, Psychosocial) and three Outcome Areas (Neurodevelopment, Obesity, Upper and Lower Airways). The ECHO Program expects each Cohort Study Site to collect all data elements in at least one Specialized Outcome Area on all participants appropriate to their age, as specified in this protocol. The ECHO Program also expects Cohort Study Sites recruiting new pregnancies to collect all data elements in at least one Specialized Exposure Area on all participants appropriate to their age, as specified in this protocol. Specialized elements in the ECHO Cohort Data and Biospecimens Collection Protocol include all of the elements in Section 4.3 below.

Term	Definition
Team Science	Team Science is a collaborative effort to address a scientific challenge that leverages the strengths and expertise of professionals trained in different fields. Team Science allows the whole to be greater than the sum of the parts by facilitating transdisciplinary research.
Transdisciplinary Research	Researchers from different disciplines work jointly to develop and use a shared conceptual framework that synthesizes and extends discipline-specific theories, concepts, and methods to create new approaches to address a common problem.
Upper and Lower Airways Outcome Domain	The outcome domain focused on upper and lower airway-related outcomes (e.g., allergy, asthma, rhinitis, sleep-disordered breathing).

1. Background and Rationale

In 2016, the NIH established the Environmental influences on Child Health Outcomes (ECHO) Program, an innovative and collaborative research initiative whose mission is to enhance the health of children for generations to come. The overarching scientific goal of ECHO is to advance understanding of the effects of a broad array of early environmental exposures on children's development and health outcomes with high public health impact. To achieve this goal, the ECHO Program will leverage and expand ECHO Cohort Study Sites that collect information on various environmental exposures, e.g., physical, chemical, social, behavioral, and biological, to support a single ECHO cohort that spawns multiple longitudinal analyses. These ECHO-supported Cohort Study Sites will employ standardized data collection methods and focus on five key outcome domains:

- pre-, peri-, and postnatal outcomes
- neurodevelopment
- upper and lower airways
- obesity and obesity-related conditions
- positive health

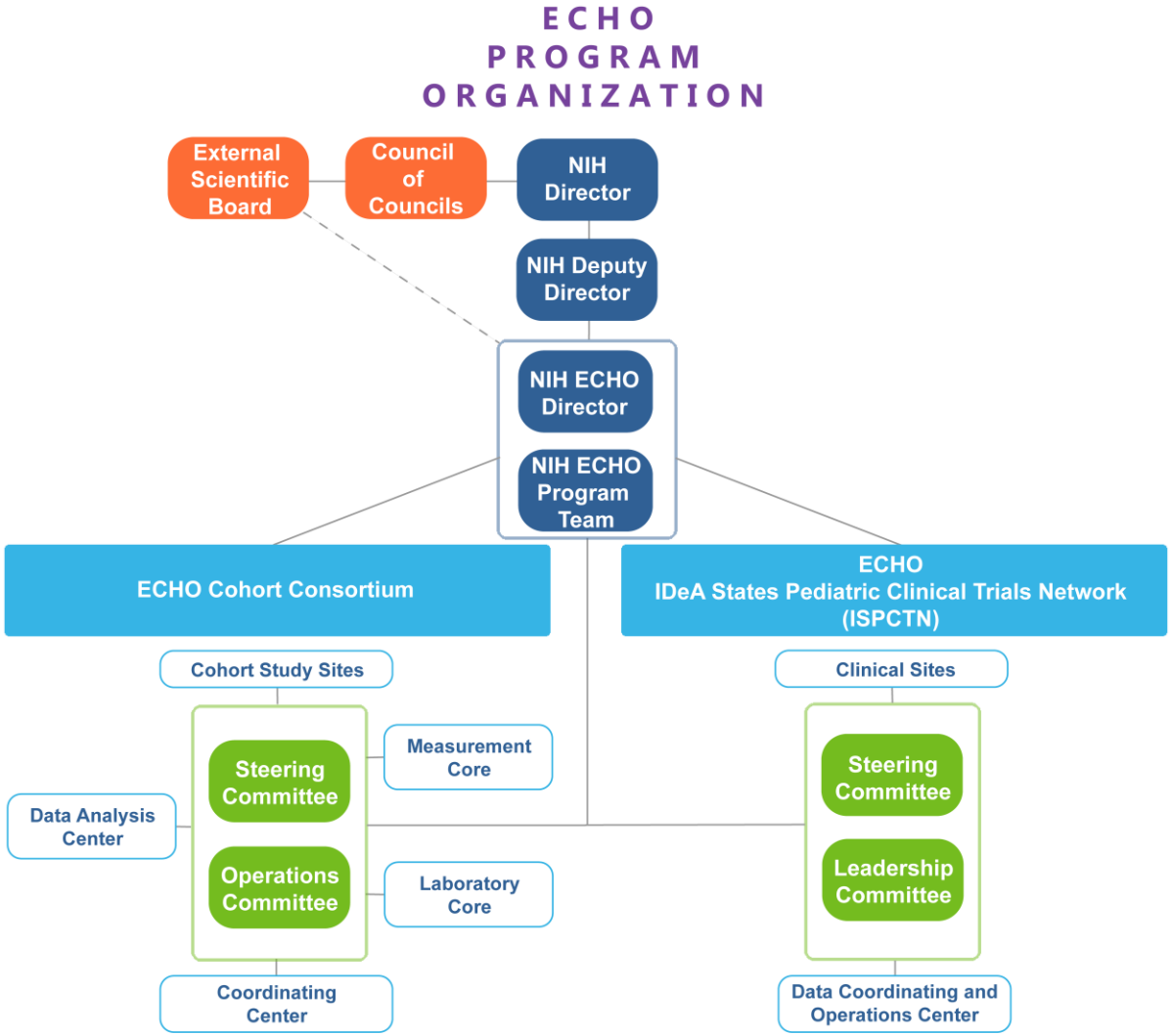
The ECHO Cohort Consortium has five components:

- the ECHO Cohort Study Sites

- the Coordinating Center (CC)
- the Data Analysis Center (DAC)
- the Measurement Core
- the Laboratory Core

Figure 1 depicts the organizational structure of the ECHO Program, including the ECHO governance structure for its two research arms - the observational ECHO Cohort Consortium and interventional IDeA States Pediatric Clinical Trials Network.

Figure 1 ECHO Program Organizational Structure


ECHO PROGRAM ORGANIZATION
External Scientific Board
Council of Councils
NIH Director
NIH Deputy Director
NIH ECHO Director
NIH ECHO Program Team
ECHO Cohort Consortium
ECHO IDeA States Pediatric Clinical Trials Network (ISPCTN)
Cohort Study Sites
Steering Committee
Measurement Core
Data Analysis Center
Operations Committee
Laboratory Core
Coordinating Center
Clinical Sites
Steering Committee
Leadership Committee
Data Coordinating and Operations Center

In Cycle 1, ECHO included 31 ECHO Cohort Awardees, comprising 69 ECHO cohorts. These existing cohorts represented diverse study populations with varying research goals, data collection procedures, measurements, and follow-up schedules, and include preconception, prenatal (over half of the ECHO Cohort were prenatal cohorts), birth, infancy, and early childhood inception cohorts. In addition to sharing extant data on their participants, these cohorts enrolled into the ECHO Cohort approximately 35,000 children and their primary caregivers for new data and biospecimen collection.

Enrollment in these cohorts spanned multiple decades, with some recruitment beginning in the early 1980s, while many other ECHO cohorts had ongoing recruitment efforts. ECHO is a nationwide program with sites distributed across the United States, including Puerto Rico, with participants of diverse race/ethnicity (including Indigenous populations) and socioeconomic status.

Cycle 2 of the ECHO Cohort Consortium, 2023 to 2030, represents the next iteration of this large NIH initiative to enhance child health. The purpose of Cycle 2 is to extend and expand the ECHO Cohort to further investigate the influences of a broad range of early exposures from society to biology, including during the preconception period, on child health.

This large consortium of Cohort Study Sites, along with ECHO's Cores and Centers, will expand the potential to reach ECHO's primary scientific goals by the formation of a single, large, and diverse ECHO Cohort.

In Cycle 2, the ECHO Cohort aims to follow the children enrolled in Cycle 1 and enroll new pregnant participants, their offspring, and, if available, the current partner. ECHO Cohort will include approximately 50,000 children plus their caregivers in Cycle 2.

Many of the pregnant participants and their partners will, after the birth of their first ECHO child, constitute a preconception cohort for a 2nd child born within Cycle 2.

2. Objective and Development of the ECHO Cohort Data and Biospecimen Collection Protocol

2.1 Primary Objective of the ECHO Cohort Data and Biospecimen Collection Protocol

The primary objective of the ECHO Cohort Data and Biospecimen Collection Protocol is to facilitate the creation of the ECHO Cohort Data Platform consisting of data elements and biospecimens from all Cohort Study Sites. This protocol serves to standardize data collection across the ECHO Program and to specify what data elements and biospecimens ECHO Cohort Study Sites should collect and share across the life course, from the prenatal period through adolescence, to create the ECHO Cohort Data Platform and the

ECHO Cohort Biorepository.

ECHO Investigators and their collaborators from the wider scientific community can use the ECHO Cohort Data Platform and the ECHO Cohort Biorepository to address innovative, high-impact, solution-oriented research questions that require multiple cohorts and relate a broad range of early environmental influences to child health outcomes.

2.2 Development of the ECHO Cohort Data and Biospecimen Collection Protocol

The ECHO Cohort Data and Biospecimen Collection Protocol is not intended to address one specific research question, but rather to facilitate the development of a data platform that investigators can use to explore numerous innovative, solution-oriented research questions across the five ECHO outcome domains.

The initial development of this protocol was a transdisciplinary, iterative process guided by the principles of Team Science. It included stakeholders across ECHO, as evidenced by close collaboration between the ECHO Protocol Development Working Group, Outcome Working Groups, the Chemical Exposures and the Biospecimens Working Groups, exposure analysis experts, the ECHO Executive Committee, ECHO Cohort Steering Committee, and the NIH ECHO Program Office.

Brief Timeline of ECHO Cohort Data and Biospecimen Collection Protocol Development

In 2016, the ECHO Program charged the Outcome Working Groups with developing broad research topics and questions to inform the development of the ECHO-wide Cohort Data Collection Protocol (later retitled, see Section 2.3). These overarching research questions were intentionally broad to stimulate discussion and were not intended to be specifically addressed by this protocol. However, these questions were vital to the protocol development process and, ultimately, to the generation of the ECHO Cohort Data Platform. For ECHO analyses relying on the ECHO Cohort Data Platform, writing groups develop analysis proposals that contain more refined research questions. The ECHO Publications Committee reviews and approves these analysis proposals. Below, in early versions of the protocol (versions 1.0 through 2.1), the childhood life course was organized across distinct developmental life stage categories. For heuristic purposes, the ECHO Program assigned specific age bands to each life stage based on the consensus of pediatric developmental experts.

The ECHO Steering Committee ratified v1.0 of the protocol in November 2017. The ECHO-wide Cohort Data Collection Protocol v1.1 was ratified by the ECHO Cohort Steering Committee on 20 July 2018. In November 2018, the Protocol Implementation and Evaluation (PIE) Committee performed a reconciliation activity between protocol Version 1.1 and all developed study materials (e.g., forms), resulting in the generation of a clarification amendment and, consequently, protocol v1.2 (dated 11 December 2018). In

August 2019, the ECHO Program activated the first ECHO cohorts under protocol v1.2. Later revisions resulted in v1.3 (approved by NIH on 09 April 2020), v2.0 (approved by NIH on 24 February 2021), and v2.1 (approved by NIH on 25 June 2021).

Section 2.3 below describes the overarching process to develop the current protocol (retitled the ECHO Cohort Data and Biospecimen Collection Protocol) from the most recent version (v2.1, titled the ECHO-wide Cohort Data Collection Protocol). Refer to Summary of Changes table in Section 10.2 for a complete list of changes made to the protocol v3.1 to the current v3.2.

2.3 Development of the ECHO Streamlined Protocol

In January 2022, the ECHO Executive Committee convened the Streamlined Protocol Task Force (SPTF, later renamed the Protocol Task Force (PTF)), composed of representatives from each of the following ECHO components: Pediatric Cohorts, DAC, Person-reported Outcomes Core, CC, and NIH ECHO Program Office. The ECHO Executive Committee charged the PTF with reviewing and modifying the existing ECHO-Wide Cohort Data Collection Protocol v2.1 to produce a streamlined version with prescribed data collection timing and measures. Therefore, the primary objective of the PTF was to create a single protocol that balanced burden with scientific value for use in NIH-funded longitudinal studies of child health. To guide the streamlining process, the PTF first established a set of guiding principles:

- Develop a protocol that supports the engagement and retention of participants of diverse backgrounds.
- Preserve scientific potential while being respectful of participant and study staff burden by setting maximum time limits for individual visits, shortening assessments, standardizing assessment within and across ages, and reducing data collection burden where possible.
- Change the frequency of data collection to emphasize more frequent visits, including several during the prenatal life stage and first two years of life, annually thereafter.
- Maximize self-administered options for remote data collection to reduce participant and staff burden and costs and increase opportunities to engage distant participants and increase retention.

The PTF relied on the previous work conducted by PIE, identified key exposure and outcome topic areas based on the existing ECHO-wide Protocol v2.1, and reviewed metadata to identify usage rates of each measure. Small groups within the PTF reviewed and suggested data elements and specific measures related to each exposure and outcome topic area.

Several features of the streamlined protocol differ from prior versions of the ECHO Cohort Protocol. First, standardizing assessments

and creating a single data collection protocol required removing the distinction between essential and recommended data elements and biospecimens as well as the options for preferred, acceptable, and alternative measures. Instead, all Cohort Study Sites must collect all data elements and biospecimens on all participants in the current protocol, and only one measure is listed per data element unless age-specific instruments are necessary.

Second, whereas prior versions used a life stage approach in which cohorts collected data and biospecimens at least once per life stage, the ECHO Cohort Data and Biospecimen Collection Protocol (ECHO Cohort Protocol) requires more frequent collection from prenatal through 35 months due to the rapid developmental changes that occur during this period, and annual data collection beginning at age 3.

Related, the ECHO Cohort Protocol includes specific age bands beginning at age 3 years, within which participants follow the same data and biospecimen collection procedures in the calendar year. While the age band approach means a proportion of children at each discrete age will provide data for any given calendar year (versus all children of a discrete age), such an approach allows for coverage of all data elements and biospecimens across all ages. The alternative design of an age-specific protocol would result in some data elements and biospecimens being collected less frequently and only at specific ages, thus leaving large gaps in assessment. Age bands were informed by the previous life stage designations and pediatric developmental experts.

In May 2022, the PTF met in-person to review small group recommendations and to develop the initial draft ECHO Cohort Protocol. The PTF then reviewed and refined the protocol draft to ensure it was cohesive and comprehensive. The draft underwent ECHO Cohort Steering Committee and ECHO Diversity, Inclusion, and Equity Task Force review and comment in July 2022. The resulting draft ECHO Cohort Data and Biospecimen Collection Protocol v3.0 was approved by the NIH ECHO Program Office on 23 August 2022.

Following additional feedback from the ECHO Cohort Consortium and the NIH ECHO Program Office, the PTF reconvened in Fall 2022 to further streamline the protocol, with an emphasis on reducing participant burden while retaining potential scientific impact. Two primary approaches to reduction were taken. First, the group identified and removed data elements, measures, and biospecimens deemed specialized and therefore not core to collect on all ECHO participants. These included highly specific data elements and biospecimens primarily relevant to only one outcome or exposure area. Second, the PTF omitted lengthy measures that could be replaced by briefer options. Third, the PTF streamlined the frequency of data collection for each individual data element and biospecimen. With few exceptions, Cohort Study Sites will collect data elements and biospecimens two to three times across the 7-year period of Cycle 2, regardless of the age at which a participant enters the protocol. Additionally, the PTF took care to ensure Cohort Study Sites collect conceptually aligned data elements and biospecimens concurrently (i.e., within the same calendar year) to

allow for meaningful analyses. A subgroup of the PIE Committee reviewed Data Collection Forms in parallel with protocol updates, requested and received input from various working groups, and revised Data Collection Forms accordingly. PIE then reviewed new and adapted Data Collection Forms with a diverse group of volunteers from the Cohorts, Cores, and Centers with age-appropriate children as well as currently or recently pregnant people. Based on the results of those reviews, PIE and the PTF made final revisions to the ECHO Cohort Protocol v3.0 and its associated forms. The final ECHO Cohort Data and Biospecimen Collection Protocol v3.0 was approved by the NIH ECHO Program Office on 25 SEP 2023.

This version of the ECHO Cohort Protocol contained core elements, i.e., elements that ECHO expects all Cohort Study Sites to collect on all participants at the appropriate ages.

The PTF then worked to design and plan preconception collections. On 21 August 2024, the NIH approved ECHO Cohort Protocol v3.1, adding current partners of previously pregnant participants and data and biospecimen collections from both previously pregnant participants and their partners during the inter-pregnancy period.

In Fall 2023, the PTF identified small subgroups of experts for each specialized area to propose data elements and specimens for inclusion in their respective specialized area. Subgroups reviewed original proposal aims from all award sites that proposed engaging in the respective specialized area to identify important elements and specimens needed to answer proposed research questions. Subgroups also reviewed the existing elements and specimens in the core protocol to understand what was already being measured. Subgroups met throughout the fall, engaging with outside experts who provided unique insight on specific measures and specimens, as needed. In December 2023, each specialized subgroup submitted a final list of data elements and specimens for their specialized area. At this time, the respective specialized subgroups confirmed that no new data elements and specimens were needed for the pre-, peri-, postnatal outcome specialized area and few new data elements for the positive health outcome. Thus neither outcome was included as a separate specialized area. Rather, the core elements in v3.1 already include all relevant PPP outcomes, and v3.2 adds a handful of positive health data elements to the protocol. The members of the aborted PPP subgroup focused on modifying the medical record abstraction forms for the protocol.

In Winter 2024, the PTF reviewed all specialized area input, which culminated in an in-person meeting in February 2024 to finalize the data elements and specimens for each specialized area of the protocol. In April 2024, the ECHO Steering Committee approved these additions.

3. ECHO Program Participant Population

3.1 Inclusion Criteria

This protocol applies to all participants, within Cohort Study Site Awards who meet one of these five criteria for an ECHO Cohort Data and Biospecimen Collection Protocol research participant:

1. Child enrolled in an ECHO Cohort Study Site* who meet the criteria: a) children born to pregnant enrollees in ECHO Cycle 2, or b) children who were enrolled for new data collection in ECHO Cycle 1.

2. Pregnant person enrolled at less than 20 weeks of pregnancy in an ECHO Cohort Study Site

3. Primary caregiver, preferably the biological mother, where possible, of the child (or children) anticipated to enroll by an ECHO Cohort Study Site. An ECHO Cohort Study Site that did not previously enroll the child's primary caregiver must seek their permission for participation under the ECHO Cohort Data and Biospecimen Collection Protocol.

4. Biological mother of a child (or children) anticipated to enroll by an ECHO Cohort Study Site, if not already enrolled under inclusion criterion #3 above, for the collection of "sample appropriate for DNA."

5. Current partner of an enrolled participant who completed an ECHO pregnancy within the prior eleven months, thirty days.

ECHO will follow children and pregnant persons for ECHO outcomes of interest. Primary caregivers (and biological mothers, as applicable) and current partners will provide valuable biospecimens and exposure/covariate/outcome-related data through self-report and proxy-report measures that will help ECHO understand how early environmental exposures influence child health outcomes.

ECHO Cohort Study Sites will seek consent (and permission/assent as appropriate) for the participation of all possible participants noted above and the collection of their data. Cohort Study Sites will also collect data from consented participants about other individuals who interact with ECHO Cohort Data and Biospecimen Collection Protocol participants. These individuals may include biological parents and siblings, non-biological parents and siblings, and other caregivers of the child, as well as partners, biological parents and siblings, and other children of the pregnant person.

3.2 Exclusion Criteria

A surrogate pregnancy is excluded from enrollment.

3.3 Duration of Participation

The expected Cycle 2 funding period for the ECHO Cohort Consortium is 7 years, starting in September 2023 (FY 23 funds). Under this cycle, Cohort Study Site will collect data through August 2030.

3.4 Number of Participants

In Cycle 1, from September 2016 to August 2023, the ECHO Program consisted of 31 Cohort Awardees that comprised 69 cohorts and their associated participating ECHO cohort sites (approximately 180 sites). In Cycle 2, from September 2023, the ECHO Program consists of 45 Cohort Study Site Awards to recruit more than 20,000 pregnant participants, with the number of children in excess of 50,000 over the course of the ECHO Program.

3.5 Participant Withdrawal

Participants can withdraw from participation (i.e., discontinue participation) in the ECHO Cohort at any time and for any reason, without consequence. In cases of participant withdrawal, ECHO study staff will record withdrawal-related information, including the reason(s) for withdrawal. Participant data and biospecimens collected prior to a participant's withdrawal will remain in the ECHO Cohort Data Platform and ECHO repository for analyses related to the ECHO Cohort Data and Biospecimen Collection Protocol (unless otherwise specified by the participant).

3.6 ECHO Cohort Data and Biospecimen Collection Protocol Design: Data Elements and Biospecimens

The ECHO Cohort Data and Biospecimens Collection Protocol serves to standardize data and biospecimen collection.

This protocol will support numerous future longitudinal analyses through the creation of the ECHO Cohort Data Platform. This data platform will include data from existing and new ECHO participants from ECHO Cohort Study Sites, who will use standardized data collection methods.

The data elements and biospecimens featured in this protocol encompass, but are not limited to, the targeted core data elements specified in the Request for Applications RFA-OD-22-017, RFA-OD-22-018 and RFA-OD-22-019, including demographics, residential address, typical early development, environmental exposures, genetics, and person-reported outcomes. The core data elements are presented in **Table 2** to **Table 8** according to the age group in which a participant falls during a specific calendar year.

The ECHO Program requires that Cohort Study Sites collect and share data elements and biospecimens; this requirement applies to all ECHO Cohort Study Sites. The ECHO Program expects that Cohort Study Sites will collect all core data elements and biospecimens from all willing participants during the period of their ECHO funding support.

Specialized Data Elements and specimens include more in-depth or frequent elements/specimens than the Core elements/specimens. They are organized into three Exposure Areas (Chemical/Physical, Lifestyle, Psychosocial) and three Outcome Areas (Neurodevelopment, Obesity, Upper and Lower Airways). The ECHO Program expects each Cohort Study Site to collect all data elements and specimens in at least one Specialized Outcome Area on all participants appropriate to their age, as specified in this protocol. The ECHO Program also expects Cohort Study Sites recruiting new pregnancies to collect all data elements and specimens in at least one Specialized Exposure Area on all participants appropriate to their age, as specified in this protocol. Specialized elements/specimens in the ECHO Cohort Data and Biospecimens Collection Protocol include all of the elements and specimens in Section 4.3 below.

The specialized data elements and specimens are presented in **Table 9** to **Table 14** by exposure and outcome area and age band, for calendar years 2025-2030.

4. ECHO Cohort Data and Biospecimen Collection Protocol Procedures

The following sections apply to data collection and describe the data elements and associated measures and biospecimens that ECHO Cohort Study Sites should collect.

The data elements and associated measures and biospecimens included in the ECHO Cohort Data and Biospecimen Collection Protocol are organized across the child developmental life course, from the prenatal period through adolescence.

The core data elements and biospecimens are presented in **Table 2** to **Table 8** according to the specific calendar year (2024 – 2030). A separate table (**Table 1**) indicates the data elements to collect at the time of the participant's first study visit, which includes obtaining informed consent. **Table 9** to **Table 14** summarize specialized data elements and specimens under Exposures and Outcomes according to the specific calendar year (2025 – 2030).

In this protocol, child self-report measures begin at age 8 years, with caregiver report versions for younger children. In some instances, both child and caregiver report measures are required for data collection within a given calendar year. If a child self-report option is available and developmentally appropriate for the cohort's specific study sample, ECHO Cohort Study Sites should use the child self-report version. However, additional considerations, such as reading ability, comprehension, and cognitive ability may necessitate primary caregiver reporting at ages older than 8 years. Caregivers may report for children above the age of assent independent of the associated child's enrollment in the ECHO Cohort Protocol. Caregiver and parent reports are valid assessments of child behavior, health, and development that yield valuable information about child health over time, even in the absence of direct child assessments.

In this version of the protocol, Cohort Study Sites administer many survey measures as computer-adaptive tests (CATs). CATs increase measurement precision with minimal burden on participants by determining which item should be administered next based on the respondent's prior answers. CATs adapt to an individual's level of the latent trait being measured, and participants can complete the CATs included in this protocol in four to six items. Measures that can be administered as a CAT are noted with the convention "/CAT" in **Table 2** to **Table 14**.

Many of the core data elements presented in the following sections can be exposures, outcomes, or covariates depending on the specific analysis; therefore, core data elements are not organized by outcome, exposure, or covariate status. For future specific analyses performed using the ECHO Cohort, writing groups will specify data elements as exposures, outcomes, or covariates in question-specific analysis proposals (see Section 6). For specialized data elements, Cohort Study Sites opt into collecting at least one specialized outcome area and for pregnancy sites, at least one specialized exposure. Therefore, specialized data elements are organized by the three exposure (Chemical/Physical, Lifestyle, Psychosocial) and three outcome (Neurodevelopment, Obesity, Upper and Lower Airways) areas.

4.1 Frequency of Data Collection

The ECHO Program expects that each Cohort Study Site will collect core data element measures and biospecimens on their cohort participants twice in the prenatal period (early pregnancy and in the 3rd trimester), in the perinatal period, and at 0 to 5 months, 6 to 11 months, 12 to 23 months, and 24 to 35 months. Beginning at age 3 years, the protocol follows a calendar-based annual visit schedule delimited into four broad child age bands: 3 to 5 years, 11 months, 30 days; 6 to 10 years, 11 months, 30 days; 11 to 17 years, 11 months, 30 days; and 18 to 20 years, 11 months, 30 days. Ages refer to the age the child turned during the calendar year of data collection.

The ECHO Program also expects that each Cohort Study Site will collect all specialized data element measures and biospecimens on all cohort participants for the specialized outcome, and if relevant, exposure, area in which the Cohort Study Site selects to participate. The specialized data element measures and biospecimens follow the same collection time points and calendar years as the core protocol, with specific measures and biospecimens outlined in **Table 9** to **Table 14**.

During a single calendar year (e.g., January 1 to December 31, 2024), Cohort Study Sites collect the same set of measures and biospecimens in a particular age band for all participants within that age band (e.g., any participant age 3, 4, or 5 years in 2024 follows the same measure and biospecimen collection procedures for 2024).

Cohort Study Sites with child participants of varying ages spanning multiple age bands should collect the set of measures and biospecimens appropriate to individual participant's age and the calendar year.

At the first point of contact with participants for Protocol v3.0 data collection, Cohort Study Sites are expected to consent/assent participants. Cohort Study Sites are also expected to collect the data elements outlined in **Table 1**: Data Elements for First Visit. Caregivers may report for children above the age of assent independent of the associated child's enrollment in the ECHO Cohort Protocol. Caregiver and parent reports are valid assessments of child behavior, health, and development that yield valuable information about child health over time, even in the absence of direct child assessments.

4.2 Core Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course

The ECHO Manual of Operating Procedures (MOP) provides detailed information regarding the data collection procedures under the ECHO Cohort Data and Biospecimen Collection Protocol. The ECHO Biospecimen Policy describes the policies and processes for the collection, processing, and storage of biospecimens.

Table 2 to **Table 8** summarize data elements under “Concepts,” which are overarching organizational constructs that encompass numerous related data elements; within concepts, data elements are generally organized chronologically across the child life course. The protocol includes the following concepts: Sociodemographics, Child and Family Health History, Adhoc Time Sensitive, Pregnancy-related, Caregiver Neuro-psychosocial, Caregiver Well-being, Built/Physical Environment, Child Physical Health, Child Health Behaviors/Lifestyle, Home Social Environment, Child Neurodevelopmental Health and Functioning, Child Social Role Performance and Functioning, Child Well-being, Child Biospecimens, Maternal and Partner Biospecimens, and Partner-related.

Biospecimens collected under the ECHO Cohort Data and Biospecimen Collection Protocol will be stored in the ECHO Cohort Biorepository. The Biorepository will be responsible for providing specimen collection kits and acquiring, processing, storing, and distributing ECHO biospecimens.

4.3 Specialized Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course

Specialized Data Elements and specimens include more in-depth or frequent elements/specimens than the Core elements/specimens. They are organized into three Exposure Areas (Chemical/Physical, Lifestyle, Psychosocial) and three Outcome Areas (Neurodevelopment, Obesity, Upper and Lower Airways). The ECHO Program expects each Cohort Study Site to collect all data elements and specimens in at least one Specialized Outcome Area on all participants appropriate to their age, as specified in this protocol. The ECHO Program also expects Cohort Study Sites recruiting new pregnancies to collect all data elements and specimens in at least one Specialized Exposure Area on all participants appropriate to their age, as specified in this protocol. Specialized elements/specimens in the ECHO Cohort Data and Biospecimens Collection Protocol include all of the elements and specimens in Section 4.3 below.

The specialized data elements and specimens are presented in **Table 9** to

Table 8 by exposure and outcome area and age band, for calendar years 2025-2030.

Table 9 to **Table 14** summarize specialized elements under Exposures and Outcomes. The protocol includes measures and biospecimens for three Exposure Areas (Chemical/Physical, Lifestyle, Psychosocial) and three Outcome Areas (Neurodevelopment, Obesity, Upper and Lower Airways).

Table 1 Data Elements for First Visit

Core Data Element	Form/Measure Name	First Visit
Concept: Sociodemographics		
Family Demographics	Demographics of Biological Family[a]	X
	Demographics of Primary Caregiver Family[a]	X
Child Demographics	Demographics of the Child[b]	X
Residential Address	Residential Address History[c]	X
Concept: Child & Family Health History		
Child Medical History	Medical History of the Child[b]	X
Concept: Caregiver Neuro-Psychosocial		
Caregiver Stressful Life Events In Pregnancy	PRAMS Stressful Life Events in Pregnancy[d]	X

[a] Cohort Study Sites are expected to collect demographic information related to the ECHO child's biological family members; however, in cases where biological family members are not the primary caregivers of the child (e.g., child is adopted, biological caregivers deceased), this information may not be available. In these cases, capturing the child's rearing family demographics is critical. If the rearing family is different than the biological family, ECHO cohorts should strive to collect both biological family and rearing family demographics to the extent possible.

[b] Cohort Study Sites starting with children >5 months of age should obtain Child Demographics and Child Medical History at the First Visit; otherwise collect these data elements at the 0-5 month visit.

[c] This measure should be completed by new ECHO participants or those who have not yet completed a residential address history in reference to the ECHO child participant who is <18 years old. Child participants who are 18-20 years old should complete this measure only if they have complete address history. Otherwise, just complete the Residential Address Update form at each visit as outlined in Tables 2-8. For child participants 18-20 years of ages, the child or the child's caregiver should complete the Residential Address Update form at the first visit.

[d] Cohort Study Sites starting with children between the ages of >5 months to 10 years of age should obtain Caregiver Stressful Life Events in Pregnancy at the First Visit; otherwise, collect in the 0-5 months age range following the birth of the child.

Table 2 Data Elements for Year 2024

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X			X	X	X	
	Income, Assistance, and Financial Strain – 18-20 yr olds											X
Caregiver Occupation	Caregiver Occupation and Employment		X			X			X	X	X	
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X		X	X		
	Language and Acculturation – 18-20yr olds											X
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Gender Identity	Gender Identity – Parent Report									X		
	Gender Identity – Child Self Report										X	X
Early Care and Education	Early Care and Education					X	X		X			
K12 Education	Child K12 Education									X	X	

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X			X	X	X	
	Health Insurance Coverage – 18-20 yr olds											X
Caregiver Health Literacy	Caregiver Health Literacy				X							
Caregiver Weight	Caregiver Weight – Self Report					X	X	X				
Concept: Adhoc Time-sensitive												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X
Concept: Pregnancy-related[e]												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT) © The Regents of the University of California, Davis campus.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/Breastfeeding	Maternal Supplements Short Form – Pregnancy[h]		X									
	Maternal Supplements Short Form – Breastfeeding[h]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X		X	X	X	
Caregiver Stressful Life Events In Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X		X	X	X	X
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X		X	X	X	X
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X		X	X	X	X
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X		X	X	X	X
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X		X	X	X	X
	PROMIS v2.0 – Informational Support 4a/CAT		X		X				X			
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X				X			
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X	X			
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X				X			

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X		X			
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-Pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X					
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-Pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X					
	Household Exposure to Secondhand Smoke – 18-20 yr olds[f]											X
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X				
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[i]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X				X
Child Blood Pressure												X

[i] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Body Composition	Bioimpedance								X[j]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[k]				X	X	X					
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[l]						X	X				
Child Substance Use	Youth Risk Behavior – Substance Use[m]											X
Child Sexual Behavior	Youth Risk Behavior – Sexual Behavior[m]											X

[j] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[k] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

[l] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[m] Adapted from the Center for Disease Control and Prevention 2017 Youth Risk Behavior Surveys (YRBS) for Middle and High School. The YRBS is part of the CDC Youth Risk Behavior Surveillance System (https://www.cdc.gov/healthyyouth/data/yrbs/index.htm).

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Home Screen Media Access and Use	Child Media Use – Parent Report[n]						X			X		
	Child Media Use – Self Report[o]									X	X	
Concept: Home Social Environment												
Child Negative Life Events	Adverse Childhood Experiences – Parent Report								X	X		
	Adverse Childhood Experiences – Child Self Report										X	
	Adverse Childhood Experiences – Adult Primary Version – 18-20 yr olds											X
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – Preschool Short Form								X			
	Alabama Parenting Questionnaire – 9-item Parent Report										X	
	Alabama Parenting Questionnaire – 9-item Child Self Report										X	

[n] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[o] Modified from the *Common Sense Census: Media Use by Tweens and Teens* Report. ©Common Sense Media Inc., 2015; *Social Media, Social Life: Teens Reveal Their Experiences, 2018* Report. ©Common Sense Media Inc., 2018; The Common Sense Census: Media Use by Tweens and Teens. ©Common Sense Media Inc., 2021.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X		X			
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT										X	
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X		X	X	X	
Neighborhood Safety	Geographic Information System (GIS)[p]	X	X		X	X	X	X	X	X	X	X
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X		X	X	X	
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Anxiety 8a/CAT – 18-20 yr olds											X

[p] Geospatial data is derived from participant residential address.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Depression 8a/CAT – 18-20 yr olds											X
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X	X			
	Child Behavior Checklist (CBCL) for 6-18 yr olds									X[q]	X	
Child Social Cognition	Social Responsiveness Scale – Second Edition (SRS-2) Preschool – Short Form								X			
	Social Responsiveness Scale – Second Edition (SRS-2) School Age – Short Form										X	
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT										X	
	NIH Toolbox v2.0 – Perceived Stress Scale 10-item – 18-20 yr olds											X
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v1.0 – Social Relationships – Peer Relationships 4a – Early Childhood Parent Report								X			
	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT										X	
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT										X	
	PROMIS v2.0 – Companionship 6a – 18-20 yr olds											X

[q] If the child is not yet 6 yr when entering the 6-10 age band, administer the CBCL 1.5-5yr instead.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Social Support	NIH Toolbox v2.0 – Emotional Support – 8-17 yr olds/CAT										X	
	NIH Toolbox v2.0 – Emotional Support – 18-20 yr olds/CAT											X
Bullying/Victimization	Healthy Pathways Bullying Scales										X	
Academic Ability	Early Academic Competencies – Parent Report[r]								X			
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X	X			
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy										X	
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric										X	
	PROMIS v1.2 – Global Health Scale – 18-20 yr olds											X
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X	X			
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X	X			
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT										X	
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT										X	
	PROMIS v1.0 – General Life Satisfaction 5a – 18-20 yr olds/CAT											X

[r] Adapted from the 2016 National Survey of Children's Health conducted by the United States Census Bureau, US Department of Health and Human Services.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Meaning And Purpose	PROMIS v1.0 – Meaning and Purpose 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Meaning and Purpose 8a – 18-20 yr olds/CAT											X
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[s]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						

[s] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Urine					X							
Child Stool					X							
Child Blood Spot					X		X					
Child Hair							X					
Child Shed Teeth[t]									X	X	X	
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[u]		X[v]							

[t] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

[u] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[v] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit.

Core Data Element for Year 2024	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[w]		X[w]					
Maternal Urine		X	X			X[w]		X[w]				
Placenta				X								
Maternal Hair					X[w]							
Concept: Partner Biospecimens[x]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[w] Collect specimens from the biological mother following end of Cycle 2 pregnancy.
[x] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 3 Data Elements for Year 2025

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X						
Caregiver Occupation	Caregiver Occupation and Employment		X			X						
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X					
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Early Care and Education	Early Care and Education					X	X					
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X
	Racial/Ethnic Identity – Child Report[c]										X	X

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b,d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X						
Caregiver Health Literacy	Caregiver Health Literacy				X							

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Caregiver Weight	Caregiver Weight – Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT)© The Regents of the University of California, Davis campus.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions				X	X	X					

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X					
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X					
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X					
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X					
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X					
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X					
	PROMIS v2.0 – Informational Support 4a/CAT		X		X							
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X							
Caregiver Stressful Life Events In Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X				
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X							
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X					

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X		X			
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X		X	X	X	
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X	X			
	Airways Questionnaire – 6-10 yr olds[i]									X		
	Airways Questionnaire – 11-20 yr olds[i]										X	
Lung Function	Spirometry									X[j]	X	

[i] This questionnaire includes PROMIS v1.0 - Asthma Impact 8a - Parent Proxy or PROMIS v1.0 - Asthma Impact 8a - Pediatric, the items of which are administered only if child has a "yes" response to the item "Does the child still have asthma?" on the Airways-related Airways Questionnaire - 6-10 yr olds or Airways Questionnaire - 11-20 yr olds.

[j] Collect Spirometry only for children age 7 and older.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[k]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X				
Child Body Composition	Bioimpedance								X[l]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[m]				X	X	X					
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[n]						X	X				
Child Substance Use	Youth Risk Behavior – Substance Use[o]										X	X

[k] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[l] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[m] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

[n] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[o] Adapted from the Center for Disease Control and Prevention 2017 Youth Risk Behavior Surveys (YRBS) for Middle and High School. The YRBS is part of the CDC Youth Risk Behavior Surveillance System (https://www.cdc.gov/healthyyouth/data/yrbs/index.htm).

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Sexual Behavior	Youth Risk Behavior – Sexual Behavior[o]										X	X
Home Screen Media Access and Use	Child Media Use – Parent Report[p]						X		X			
Concept: Home Social Environment												
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – 9-item Parent Report									X		
	Alabama Parenting Questionnaire – 9-item Child Self Report									X		
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X					
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT									X		
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X					
Neighborhood Safety	Geographic Information System (GIS)[q]	X	X		X	X	X	X	X	X	X	X
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X					
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				

[p] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[q] Geospatial data is derived from participant residential address.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Discrimination Distress	Adolescent Discrimination Distress Index (ADDI)										X	X
Child Religiosity and Spirituality	Religiosity and Spirituality – Child Report										X	X
Child Positive Life Events	Benevolent Childhood Experiences – Child Report										X	
	Benevolent Childhood Experiences – 18-20yr olds											X
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child General Cognition	NIH Toolbox Early Cognition Battery V3								X			
	NIH Toolbox Cognition Battery V3									X	X	
	Mobile Toolbox Cognition Battery											X
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
	Rothbart Childhood Behavior Questionnaire Very Short Form								X			
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT									X		
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT									X		
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X				

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Social Cognition	Social Responsiveness Scale – Second Edition (SRS-2) School Age – Short Form									X		
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT									X		
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT									X		
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT									X		
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X				
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy									X		
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric									X		
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X				
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X				
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT									X		
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT									X		
Child Self Worth	Harter Global Self-Worth Scale – Children									X		
	Harter Global Self-worth scale – Adolescents										X	X

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X	X			
	PROMIS v1.0 – Positive Affect – Parent Proxy 4a/CAT									X		
	PROMIS v1.0 – Positive Affect – Pediatric 4a/CAT									X	X	
	PROMIS v1.0 – Positive Affect 4a/CAT											X
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[r]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						

[r] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Child Saliva for DNA									X			
Cord Blood				X								
Child Urine					X				X	X	X	
Child Stool					X							
Child Blood Spot					X		X		X			X
Child Whole Blood										X	X	
Child Hair							X					
Child Shed Teeth[s]									X	X	X	

[s] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

Core Data Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[t]		X[u]							
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[v]		X[v]					
Maternal Urine		X	X			X[v]		X[v]				
Placenta				X								
Maternal Hair					X[v]							
Concept: Partner Biospecimens[w]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[t] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[u] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit

[v] Collect specimens from the biological mother following end of Cycle 2 pregnancy.

[w] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 4 Data Elements for Year 2026

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X						
Caregiver Occupation	Caregiver Occupation and Employment		X			X						
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X					
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Early Care and Education	Early Care and Education					X	X					
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b,d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X						
Caregiver Health Literacy	Caregiver Health Literacy				X							

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Weight	Caregiver Weight- Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT) © The Regents of the University of California, Davis campus.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Maternal Medical Record Abstraction				X							
	Pregnancy Medical Conditions and Interventions[f]					X						
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X					
Caregiver Stressful Life Events in Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X					
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X					
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X					
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X					
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X					
	PROMIS v2.0 – Informational Support 4a/CAT		X		X							
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X							
Caregiver Cognitive Functioning	Mobile Toolbox Cognition Battery								X	X	X	X

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X				
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X							
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X					
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X					
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X					
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X				
	Airways Questionnaire – 11-20 yr olds[i]											X
Lung Function	Spirometry											X

[i] This questionnaire includes PROMIS v1.0 - Asthma Impact 8a - Parent Proxy or PROMIS v1.0 - Asthma Impact 8a - Pediatric, the items of which are administered only if child has a “yes” response to the item “Does the child still have asthma?” on the Airways-related Airways Questionnaire - 6-10 yr olds or Airways Questionnaire - 11-20 yr olds.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[j]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X	X	X	X	
Child Blood Pressure									X[k]	X	X	
Child Body Composition	Bioimpedance								X[l]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[m]				X	X	X					
Child Diet	Dietary Screener Questionnaire – Parent Report[h]								X	X		
	Dietary Screener Questionnaire – Self Report[h]											X
Endurance/Fitness	Step Test									X		X

[j] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[k] In the 3-5 yrs old age band, only measure Blood Pressure for children 3 yrs old and older. Do not measure Blood Pressure on a 2 year old child who is turning 3 in the calendar year.

[l] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[m] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Device-Measured Physical Activity, Sleep	Accelerometer								X	X		X
	Sleep/Physical Activity Diary – Parent Report								X	X		
	Sleep/Physical Activity Diary – Child Self Report											X
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[n]						X	X	X	X		
	Sleep Health of Adults – 18-20 yr olds[g]											X
Home Screen Media Access and Use	Child Media Use – Parent Report[o]						X					
	Child Media Use – Self Report[p]											X

[n] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[o] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[p] Modified from the *Common Sense Census: Media Use by Tweens and Teens* Report. ©Common Sense Media Inc., 2015; *Social Media, Social Life: Teens Reveal Their Experiences, 2018* Report. ©Common Sense Media Inc., 2018; The Common Sense Census: Media Use by Tweens and Teens. ©Common Sense Media Inc., 2021.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Home Social Environment												
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – 9-item Parent Report										X	
	Alabama Parenting Questionnaire – 9-item Child Self Report										X	
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X					
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT										X	
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X					
Neighborhood Safety	Geographic Information System (GIS)[q]	X	X		X	X	X	X	X	X	X	X
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X					
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				

[q] Geospatial data is derived from participant residential address.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT										X	
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT										X	
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X	X			
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT										X	
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT										X	
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT										X	
Child Social Support	NIH Toolbox v2.0 – Emotional Support – 8-17 yr olds/CAT										X	
Bullying/Victimization	Healthy Pathways Bullying Scales										X	
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X				
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy										X	
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric										X	
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X				

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X				
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT										X	
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT										X	
Child Meaning And Purpose	PROMIS v1.0 – Meaning and Purpose 8a – Pediatric/CAT										X	
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X				
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[r]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						

[r] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Urine					X							
Child Stool					X							
Child Blood Spot					X		X					
Child Hair							X				X	
Child Shed Teeth[S]									X	X	X	

[S] Teeth collected will be shed deciduous (baby) teeth. There is no “age limit” for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it’s been since it has fallen out.

Core Data Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[t]		X[u]							
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[v]		X[v]					
Maternal Urine		X	X			X[v]		X[v]				
Placenta				X								
Maternal Hair					X[v]							
Concept: Partner Biospecimens[w]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[t] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[u] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit

[v] Collect specimens from the biological mother following end of Cycle 2 pregnancy.

[w] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 5 Data Elements for Year 2027

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X			X	X	X	
	Income, Assistance, and Financial Strain – 18-20 yr olds											X
Caregiver Occupation	Caregiver Occupation and Employment		X			X			X	X	X	
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X		X	X	X	
	Language and Acculturation – 18-20yr olds											X
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Gender Identity	Gender Identity – Parent Report									X		
	Gender Identity – Child Self Report											X
Early Care and Education	Early Care and Education					X	X		X			
K12 Education	Child K12 Education									X	X	

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X
	Racial/Ethnic Identity – Child Report[c]											X
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b,d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X			X	X	X	
	Health Insurance Coverage – 18-20 yr olds											X
Caregiver Health Literacy	Caregiver Health Literacy				X							
Caregiver Weight	Caregiver Weight-Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT) © The Regents of the University of California, Davis campus.

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X		X	X	X	
Caregiver Stressful Life Events in Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X		X	X	X	X

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X		X	X	X	X
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X		X	X	X	X
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X		X	X	X	X
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X		X	X	X	X
	PROMIS v2.0 – Informational Support 4a/CAT		X		X				X			
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X				X			
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X	X			
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X				X			
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X		X			
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X					
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X					
	Household Exposure to Secondhand Smoke – 18-20 yr olds[f]											X

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X				
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[i]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X				X
Child Blood Pressure												X
Child Body Composition	Bioimpedance								X[j]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[k]				X	X	X					
Child Diet	Dietary Screener Questionnaire – Self Report[h]										X	
Endurance/Fitness	Step Test										X	

[i] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[j] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[k] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Device-Measured Physical Activity, Sleep	Accelerometer										X	
	Sleep/Physical Activity Diary – Child Self Report										X	
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[l]						X	X				
	Sleep Health of Children and Adolescents – Child Self Report[m]										X	
Child Substance Use	Youth Risk Behavior – Substance Use[n]										X	X
Child Sexual Behavior	Youth Risk Behavior – Sexual Behavior[n]										X	X
Home Screen Media Access and Use	Child Media Use – Parent Report[o]						X		X			

[l] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[m] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a - Pediatric and PROMIS v1.0 - Sleep-Related Impairment 4a - Pediatric.

[n] Adapted from the Center for Disease Control and Prevention 2017 Youth Risk Behavior Surveys (YRBS) for Middle and High School. The YRBS is part of the CDC Youth Risk Behavior Surveillance System (https://www.cdc.gov/healthyyouth/data/yrbs/index.htm).

[o] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Home Social Environment												
Child Negative Life Events	Adverse Childhood Experiences – Parent Report								X	X		
	Adverse Childhood Experiences – Child Self Report										X	
	Adverse Childhood Experiences Adult Primary Version – 18-20 yr olds											X
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – Preschool Short Form								X			
	Alabama Parenting Questionnaire – 9-item Parent Report									X		
	Alabama Parenting Questionnaire – 9-item Child Self Report									X		
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X		X			
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT									X		
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X		X			
Neighborhood Safety	Geographic Information System (GIS)[p]	X	X		X	X	X	X	X	X	X	X
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X		X	X	X	
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				

[p] Geospatial data is derived from participant residential address.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Discrimination Distress	Adolescent Discrimination Distress Index (ADDI)											X
Child Religiosity and Spirituality	Religiosity and Spirituality – Child Report											X
Child Positive Life Events	Benevolent Childhood Experiences – 18-20yr olds											X
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT									X		
	PROMIS v1.0 – Anxiety 8a/CAT – 18-20 yr olds											X
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT									X		
	PROMIS v1.0 – Depression 8a/CAT – 18-20 yr olds											X
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X				
	Child Behavior Checklist (CBCL) for 6-18 yr olds									X[q]	X	

[q] If the child is not yet 6 when entering the 6-10 age band, administer the CBCL 1.5-5yr instead.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Social Cognition	Social Responsiveness Scale – Second Edition (SRS-2) Preschool – Short Form								X			
	Social Responsiveness Scale – Second Edition (SRS-2) School Age – Short Form									X	X	
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT									X		
	NIH Toolbox v2.0 – Perceived Stress Scale 10-item – 18-20 yr olds											X
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v1.0 – Social Relationships – Peer Relationships 4a – Early Childhood Parent Report								X			
	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT									X		
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT									X		
	PROMIS v2.0 – Companionship 6a – 18-20 yr olds											X
Child Social Support	NIH Toolbox v2.0 – Emotional Support – 18-20 yr olds/CAT											X
Academic Ability	Early Academic Competencies – Parent Report[r]								X			

[r] Adapted from the 2016 National Survey of Children's Health conducted by the United States Census Bureau, US Department of Health and Human Services.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X	X			
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy									X		
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric									X		
	PROMIS v1.2 – Global Health Scale – 18-20 yr olds											X
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X	X			
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X	X			
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT									X		
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT									X		
	PROMIS v1.0 – General Life Satisfaction 5a- 18-20 yr olds/CAT											X
Child Meaning And Purpose	PROMIS v1.0 – Meaning and Purpose 8a – 18-20 yr olds/CAT											X
Child Self-Worth	Harter Global Self-Worth Scale – Children									X		
	Harter Global Self-Worth Scale – Adolescent											X

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X	X			
	PROMIS v1.0 – Positive Affect – Parent Proxy 4a/CAT									X		
	PROMIS v1.0 – Positive Affect – Pediatric 4a/CAT									X		
	PROMIS v1.0 – Positive Affect 4a/CAT											X
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[s]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						

[s] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Urine					X							
Child Stool					X							
Child Blood Spot					X		X					
Child Hair							X		X	X		
Child Shed Teeth[t]									X	X	X	

[t] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

Core Data Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[u]		X[v]							
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[w]		X[w]					
Maternal Urine		X	X			X[w]		X[w]				
Placenta				X								
Maternal Hair					X[w]							
Concept: Partner Biospecimens[x]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[u] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[v] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit.

[w] Collect specimens from the biological mother following end of Cycle 2 pregnancy.

[x] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 6 Data Elements for Year 2028

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X						
Caregiver Occupation	Caregiver Occupation and Employment		X			X						
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X					
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Early Care and Education	Early Care and Education					X	X					
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X
	Racial/Ethnic Identity – Child Report[c]										X	

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b, d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X						
Caregiver Health Literacy	Caregiver Health Literacy				X							

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Weight	Caregiver Weight-Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT) © The Regents of the University of California, Davis campus.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X					
Caregiver Stressful Life Events in Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X					
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X					
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X					
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X					
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X					
	PROMIS v2.0 – Informational Support 4a/CAT		X		X							
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X							
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X				

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X							
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X					
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X		X			
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X		X	X	X	
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X	X			
	Airways Questionnaire – 6-10 yr olds[i]									X		
	Airways Questionnaire – 11-20 yr olds[i]										X	
Lung Function	Spirometry									X[j]	X	

[i] This questionnaire includes PROMIS v1.0 - Asthma Impact 8a - Parent Proxy or PROMIS v1.0 - Asthma Impact 8a - Pediatric, the items of which are administered only if child has a "yes" response to the item "Does the child still have asthma?" on the Airways-related Airways Questionnaire - 6-10 yr olds or Airways Questionnaire - 11-20 yr olds.

[j] Collect Spirometry only for children age 7 and older.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[k]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X				
Child Body Composition	Bioimpedance								X[l]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[m]				X	X	X					
Child Diet	Dietary Screener Questionnaire – Parent Report[h]									X		
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[n]						X	X	X	X		

[k] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[l] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[m] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

[n] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Home Screen Media Access and Use	Child Media Use – Parent Report[o]						X			X		
	Child Media Use – Self Report[p]									X	X	
Concept: Home Social Environment												
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – 9-item Parent Report										X	
	Alabama Parenting Questionnaire – 9-item Child Self Report										X	
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X					
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT										X	
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X					
Neighborhood Safety	Geographic Information System (GIS)[q]	X	X		X	X	X	X	X	X	X	X

[o] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[p] Modified from the *Common Sense Census: Media Use by Tweens and Teens* Report. ©Common Sense Media Inc., 2015; *Social Media, Social Life: Teens Reveal Their Experiences, 2018* Report. ©Common Sense Media Inc., 2018; The Common Sense Census: Media Use by Tweens and Teens. ©Common Sense Media Inc., 2021.

[q] Geospatial data is derived from participant residential address.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X					
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				
Child Discrimination Distress	Adolescent Discrimination Distress Index (ADDI)										X	
Child Religiosity and Spirituality	Religiosity and Spirituality – Child Report										X	
Child Positive Life Events	Benevolent Childhood Experiences – Child Report										X	
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child General Cognition	NIH Toolbox Early Cognition Battery V3								X			
	Mobile Toolbox Cognition Battery											X
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
	Rothbart Childhood Behavior Questionnaire Very Short Form								X			
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT										X	
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT										X	
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X	X			

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT										X	
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT										X	
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT										X	
Child Social Support	NIH Toolbox v2.0 – Emotional Support – 8-17 yr olds/CAT										X	
Bullying/Victimization	Healthy Pathways Bullying Scales										X	
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X				
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy										X	
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric										X	
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X				
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X				
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT										X	
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT										X	
Child Meaning And Purpose	PROMIS v1.0 – Meaning and Purpose 8a – Pediatric/CAT										X	

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Self-Worth	Harter Global Self-Worth Scale – Adolescents										X	
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X				
	PROMIS v1.0 – Positive Affect – Pediatric 4a/CAT										X	
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[r]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						
Partner Discrimination	The Everyday Discrimination Scale					X						

[r] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Saliva for DNA									X			
Child Urine					X				X	X	X	
Child Stool					X							
Child Blood Spot					X		X		X			X
Child Whole Blood										X	X	
Child Hair							X					
Child Shed Teeth[s]									X	X	X	

[s] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

Core Data Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[t]		X[u]							
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[v]		X[v]					
Maternal Urine		X	X			X[v]		X[v]				
Placenta				X								
Maternal Hair					X[v]							
Concept: Partner Biospecimens[w]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[t] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[u] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit.

[v] Collect specimens from the biological mother following end of Cycle 2 pregnancy.

[w] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 7 Data Elements for Year 2029

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X						
Caregiver Occupation	Caregiver Occupation and Employment		X			X						
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X					
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Early Care and Education	Early Care and Education					X	X					
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b,d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X						
Caregiver Health Literacy	Caregiver Health Literacy				X							

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Weight	Caregiver Weight-Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT) © The Regents of the University of California, Davis campus.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				

[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X					
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X					
Caregiver Stressful Life Events in Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X					
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X					
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X					
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X					
	PROMIS v2.0 – Informational Support 4a/CAT		X		X							
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X							
Caregiver Cognitive Functioning	Mobile Toolbox Cognition Battery								X			
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X				

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X							
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X					
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X					
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X					
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X				
	Airways Questionnaire – 11-20 yr olds[i]											X
Lung Function	Spirometry											X

[i] This questionnaire includes PROMIS v1.0 - Asthma Impact 8a - Parent Proxy or PROMIS v1.0 - Asthma Impact 8a - Pediatric, the items of which are administered only if child has a “yes” response to the item “Does the child still have asthma?” on the Airways-related Airways Questionnaire - 6-10 yr olds or Airways Questionnaire - 11-20 yr olds.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[j]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X	X			
Child Blood Pressure									X[k]			
Child Body Composition	Bioimpedance								X[l]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[m]				X	X	X					
Child Diet	Dietary Screener Questionnaire – Parent Report[h]								X			
	Dietary Screener Questionnaire – Self Report[h]											X
Endurance/Fitness	Step Test											X

[j] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[k] In the 3-5 yrs old age band, only measure Blood Pressure for children 3 yrs old and older. Do not measure Blood Pressure on a 2 year old child who is turning 3 in the calendar year.

[l] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

[m] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Device-Measured Physical Activity, Sleep	Accelerometer								X			X
	Sleep/Physical Activity Diary – Parent Report								X			
	Sleep/Physical Activity Diary – Child Self Report											X
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[n]						X	X				
	Sleep Health of Adults – 18-20 yr olds[g]											X
Child Substance Use	Youth Risk Behavior – Substance Use[o]										X	X
Child Sexual Behavior	Youth Risk Behavior – Sexual Behavior[o]										X	X
Home Screen Media Access and Use	Child Media Use – Parent Report[p]						X					
	Child Media Use – Self Report[q]											X

[n] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[o] Adapted from the Center for Disease Control and Prevention 2017 Youth Risk Behavior Surveys (YRBS) for Middle and High School. The YRBS is part of the CDC Youth Risk Behavior Surveillance System (https://www.cdc.gov/healthyyouth/data/yrbs/index.htm).

[p] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[q] Modified from the *Common Sense Census: Media Use by Tweens and Teens* Report. ©Common Sense Media Inc., 2015; *Social Media, Social Life: Teens Reveal Their Experiences, 2018* Report. ©Common Sense Media Inc., 2018; The Common Sense Census: Media Use by Tweens and Teens. ©Common Sense Media Inc., 2021.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Home Social Environment												
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – 9-item Parent Report									X	X	
	Alabama Parenting Questionnaire – 9-item Child Self Report									X	X	
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X					
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT									X		
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X			X	X	
Neighborhood Safety	Geographic Information System (GIS)[r]	X	X		X	X	X	X	X	X	X	X
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X			X		
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child General Cognition	NIH Toolbox Cognition Battery V3									X	X	

[r] Geospatial data is derived from participant residential address.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT									X		
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT									X		
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X				
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT									X		
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT									X		
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT									X		
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT									X		
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X				
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy									X		
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric									X		
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X				

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Flexibility	PROMIS v1.0 – Self Regulation – Flexibility 5a – Early Childhood Parent Report							X				
Child Life Satisfaction	PROMIS v1.0 – Life Satisfaction 8b – Parent Proxy/CAT									X		
	PROMIS v1.0 – Life Satisfaction 8b – Pediatric/CAT									X		
Child Self-Worth	Harter Global Self-Worth Scale – Children									X		
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X				
	PROMIS v1.0 – Positive Affect – Parent Proxy 4a/CAT									X		
	PROMIS v1.0 – Positive Affect – Pediatric 4a/CAT									X		
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[S]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						

[S] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Urine					X							
Child Stool					X							
Child Blood Spot					X		X					
Child Hair							X			X	X	

Core Data Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Shed Teeth[t]									X	X	X	
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[u]		X[v]							
Maternal Whole Blood		X	X									
Maternal Blood Spot					X[w]		X[w]					
Maternal Urine		X	X			X[w]		X[w]				
Placenta				X								
Maternal Hair					X[w]							
Concept: Partner Biospecimens[x]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[t] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

[u] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[v] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit.

[w] Collect specimens from the biological mother following end of Cycle 2 pregnancy.

[x] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 8 Data Elements for Year 2030

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Sociodemographics												
Household Income, Receipt of Federal Assistance, Financial Strain	Income, Assistance, and Financial Strain – Pregnancy	X										
	Income, Assistance, and Financial Strain – Childhood					X			X	X	X	
	Income, Assistance, and Financial Strain – 18-20 yr olds											X
Caregiver Occupation	Caregiver Occupation and Employment		X			X			X	X	X	
Household Primary Language Spoken, Read	Language and Acculturation – Pregnancy	X										
	Language and Acculturation – Childhood						X		X	X	X	
	Language and Acculturation – 18-20 yr olds											X
Residential Address	Residential Address Update		X			X	X	X	X	X	X	X
Gender Identity	Gender Identity – Child Self Report										X	X
Early Care and Education	Early Care and Education					X	X		X			
K12 Education	Child K12 Education									X	X	

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Racial and Ethnic Identity	Racial/Ethnic Identity[b,c]	X	X		X	X	X	X	X	X	X	X
	Racial/Ethnic Identity – Child Report[c]										X	X
Skin Tone Assessment	Skin Tone Assessment[b,d]	X	X		X	X	X	X	X	X	X	X
	Skin Tone Assessment – Parent Report[b,d]				X	X	X	X	X	X		
	Skin Tone Assessment – Child Report[b,d]									X	X	X
Parental Leave and Service Use	Parental Leave and Service Use						X					
Concept: Child and Family Health History												
Child Medical History	Medical History of the Child Update					X	X	X	X	X	X	
	Medical History of the Child Update – 18-20 yr olds											X
Lifetime/Updated Family Medical Conditions	Medical History of the Family	X			X	X	X	X	X	X	X	X

[b] Collect this data element only once over the course of the ECHO Study. Collect this data element at the first visit, or, if the first visit already occurred, collect it at the next visit.

[c] The Racial/Ethnic Identity measure is based on the Multigroup Ethic Identity Measure (MIEM): Roberts, R. E., Phinney, J. S., Masse, L. C., Chen, Y. R., Roberts, C. R., & Romero, A. (1999). The structure of ethnic identity in young adolescents from diverse ethnocultural groups. Journal of Early Adolescence, 19(3), 301–322.

[d] The Skin Tone Assessment is based on the Yadon-Ostfeld Skin Color Scale: Ostfeld MC, Yadon ND. ¿ Mejorando La Raza?: The political undertones of Latinos' skin color in the United States. Social Forces. 2022 Jun;100(4):1806-32; Yadon N. "They Say We're Violent": The Multidimensionality of Race in Perceptions of Police Brutality and BLM. Perspectives on Politics. 2022 Dec;20(4):1209-25.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Childbirth / Neonatal Information	Childbirth / Neonatal Medical Record Abstraction				X							
	Childbirth / Neonatal Medical Record Abstraction – Section J Addendum				X							
	Child Birth Information					X						
Caregiver and Child Health Insurance Status	Health Insurance Coverage	X				X			X	X	X	
	Health Insurance Coverage – 18-20 yr olds											X
Caregiver Health Literacy	Caregiver Health Literacy				X							
Caregiver Weight	Caregiver Weigh-Self Report					X	X	X				
Concept: Adhoc Time-sensitive[e]												
Ad hoc Time-sensitive Data Collection	Ad hoc Time-sensitive Data Collection – Caregiver Report		X			X	X	X	X	X	X	
	Ad hoc Time-sensitive Data Collection – Child Report										X	X

[e] This concept represents the possibility for a future national issue or event (e.g., COVID-19 pandemic) that the ECHO Program deems relevant for the study of child health and thus should be evaluated by the ECHO Cohort Data and Biospecimen Collection Protocol. Ad hoc time-sensitive measures may also include assessments of participant feedback.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Pregnancy-related												
Maternal Substance Use In Pregnancy	Lifestyle – Early Pregnancy[f]	X										
	Lifestyle – 3rd Trimester[f]		X									
	Lifestyle – Substance Use Recall[f]				X							
Maternal Weight and Height	Maternal Weight and Height – Measured	X	X		X							
Pre-pregnancy Weight and Weight Gain During Pregnancy	Maternal Medical Record Abstraction				X							
Medical Records - Maternal Medications During Pregnancy	Maternal Medical Record Abstraction – Section G Addendum				X							
Medical Records - Maternal Medications During Labor and Delivery	Maternal Medical Record Abstraction – Section K Addendum				X							
Maternal Pregnancy-Related Conditions, Medications, etc.	Pregnancy Medical Conditions and Interventions[f]					X						
	Maternal Medical Record Abstraction				X							
Maternal Sleep Health in Pregnancy	Sleep Health in Pregnancy[g]	X	X									

[f] Adapted from The Early Life Exposures Assessment Tool (ELEAT)© The Regents of the University of California, Davis campus.
[g] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a and PROMIS v1.0 - Sleep-Related Impairment 4a.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Maternal food Contaminants in Pregnancy	Maternal Food Source and Preparation		X			X						
Maternal Supplements In Pregnancy/ Breastfeeding	Maternal Supplements Short Form – Pregnancy[f]		X									
	Maternal Supplements Short Form – Breastfeeding[f]				X							
Maternal Diet in Pregnancy	NCI Diet History Questionnaire III	X										
	Dietary Screener Questionnaire – Self Report[h]		X			X		X				
Device-Measured Physical Activity, Sleep In Pregnancy	Accelerometer	X										
	Sleep/Physical Activity Diary – Pregnant Person	X										
Pregnancy Status and Intentions	Pregnancy Intentions					X	X	X				
Concept: Caregiver Neuro-psychosocial												
Caregiver Stressful Life Events In Childhood and Adulthood	Life Stressor Checklist – Revised (LSC-R)	X										
Caregiver Stressful Life Events In Adulthood	Crisis in the Family Systems – Revised (CRISYS-R) Short Form		X				X		X	X	X	
Caregiver Discrimination	The Everyday Discrimination Scale		X		X		X		X	X	X	X

[h] Adapted from the National Health and Nutrition Examination Survey (NHANES) Dietary Screener Questionnaire (DSQ) 2009-10.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver Stressful Life Events in Childhood	Adverse Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Positive Life Events in Childhood	Benevolent Childhood Experiences[b]	X	X		X	X	X	X	X	X	X	X
Caregiver Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT		X		X		X		X	X	X	X
Caregiver Anxiety	PROMIS v1.0 Anxiety 8a/CAT		X		X		X		X	X	X	X
Caregiver Perceived Stress	NIH Toolbox v2.0 – Perceived Stress Scale 10-item		X		X		X		X	X	X	X
Caregiver Social Support	PROMIS v2.0 – Emotional Support 4a/CAT		X		X		X		X	X	X	X
	PROMIS v2.0 – Informational Support 4a/CAT		X		X				X			
	PROMIS v2.0 – Instrumental Support 4a/CAT		X		X				X			
Concept: Caregiver Well-being												
Caregiver Global Health	PROMIS v1.2 – Global Health Scale		X			X		X	X			
Caregiver Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a/CAT				X				X			
Caregiver Sleep Health	Sleep Health of Adults[g]				X		X		X			
Concept: Built/Physical Environment												
Chemical Exposures	Household Chemical Exposures – Pre-pregnancy and Pregnancy[f]		X									
	Household Chemical Exposures – Current[f]						X		X			

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Household Exposure to Secondhand Smoke	Household Exposure to Secondhand Smoke – Pre-pregnancy and Pregnancy[f]		X									
	Household Exposure to Secondhand Smoke – Current[f]				X		X		X	X		
	Household Exposure to Secondhand Smoke – 18-20 yr olds[f]											X
Concept: Child Physical Health												
Airways-related	Airways Questionnaire – 6-11 mo olds					X						
	Airways Questionnaire – 1-5 yr olds							X				
Child Height (Length)/ Weight	Child Anthropometry and Physical Examination[i]				X	X	X	X	X	X	X	X
Child Head Circumference					X			X				
Child Waist Circumference								X		X	X	X
Child Blood Pressure										X	X	X
Child Body Composition	Bioimpedance								X[j]	X	X	X
Child Pubertal Development	Pubertal Development Scale									X	X	

[i] There are several child anthropometry and physical examination forms for use dependent on the participant age and applicable measure(s). Please refer to the Child Anthropometry and Physical Examination MIS located in the MOP for details.

[j] In the 3-5 yrs old age band, measure bioimpedance only for children 5 yrs old and older. Do not measure bioimpedance on a 4-year-old child who is turning 5 in the calendar year. There are limited, market-ready devices validated for children <5yrs.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Child Health Behaviors/Lifestyle												
Infant Feeding and Introduction to Solids	Infant Feeding Practices[k]				X	X	X					
Child Diet	Dietary Screener Questionnaire – Parent Report[h]									X		
	Dietary Screener Questionnaire – Self Report[h]										X	
Endurance/Fitness	Step Test									X	X	
Child Device-Measured Physical Activity, Sleep	Accelerometer									X	X	
	Sleep/Physical Activity Diary – Parent Report									X		
	Sleep/Physical Activity Diary – Child Self Report										X	
Child Sleep Health	Sleep Health of Infants				X	X						
	Sleep Health of Children and Adolescents – Parent Report[l]						X	X	X	X		
	Sleep Health of Children and Adolescents – Child Self Report[m]										X	
Child Substance Use	Youth Risk Behavior – Substance Use[n]											X

[k] Adapted from the Infant Feeding Practices Study II postnatal questionnaires available at http://www.cdc.gov/breastfeeding/data/ifps/questionnaires.htm.

[l] This questionnaire includes PROMIS v1.0 - Sleep Problems - Disturbance 4a - Early Childhood Parent Report, PROMIS v1.0 - Sleep Disturbance 4a - Parent Proxy, and PROMIS v1.0 - Sleep-Related Impairment 4a - Parent Proxy.

[m] This questionnaire includes PROMIS v1.0 - Sleep Disturbance 4a - Pediatric and PROMIS v1.0 - Sleep-Related Impairment 4a - Pediatric.

[n] Adapted from the Center for Disease Control and Prevention 2017 Youth Risk Behavior Surveys (YRBS) for Middle and High School. The YRBS is part of the CDC Youth Risk Behavior Surveillance System (https://www.cdc.gov/healthyyouth/data/yrbs/index.htm).

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Sexual Behavior	Youth Risk Behavior – Sexual Behavior[n]											X
Home Screen Media Access and Use	Child Media Use – Parent Report[o]						X		X			
Concept: Home Social Environment												
Child Negative Life Events	Adverse Childhood Experiences – Parent Report								X	X		
	Adverse Childhood Experiences – Child Self Report										X	
	Adverse Childhood Experiences – Adult Primary Version – 18-20 yr olds											X
Caregiving Quality and Behavior	Alabama Parenting Questionnaire – Toddler Short Form						X					
	Alabama Parenting Questionnaire – Preschool Short Form								X			
Family Relationships	PROMIS v1.0 – Social Relationships – Family Relationships 4a – Early Childhood Parent Report						X		X			
	PROMIS v1.0 – Family Relationships 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Family Relationships 4a – Pediatric/CAT										X	
Neighborhood Capital	Neighborhood Collective Efficacy – Caregiver						X		X			
Neighborhood Safety	Geographic Information System (GIS)[p]	X	X		X	X	X	X	X	X	X	X

[o] Modified from Zero to Eight: Children's Media in America 2013 Report and Common Sense Census: Media Use by Kids Ages Zero to Eight Report. ©Common Sense Media Inc., 2013; 2017; 2020; The Common Sense Census: Plugged-In Parents of Tweens and Teens Report ©Common Sense Media Inc., 2016.

[p] Geospatial data is derived from participant residential address.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Caregiver-Partner Relationship Quality	Couples Satisfaction Index – 4-item Short Form	X			X		X		X			
Caregiver Substance Use	Lifestyle – Current[f]					X	X	X				
Child Discrimination Distress	Adolescent Discrimination Distress Index (ADDI)										X	X
Child Religiosity and Spirituality	Religiosity and Spirituality – Child Report										X	X
Child Positive Life Events	Benevolent Childhood Experiences – Child Report										X	
	Benevolent Childhood Experiences – 18-20yr olds											X
Concept: Child Neurodevelopmental Health and Functioning												
Child Developmental Milestones	Ages and Stages Questionnaire					X		X				
Child Temperament	Rothbart Infant Behavior Questionnaire Revised Very Short Form					X						
	Rothbart Early Childhood Behavior Questionnaire Very Short Form							X				
Child ASD screener	Modified Checklist for Autism in Toddlers Revised (M-CHAT-R)							X				
Child Anxiety	PROMIS v2.0 – Anxiety 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Anxiety 8a/CAT – 18-20 yr olds											X
Child Depressive Symptoms	PROMIS v2.0 – Depressive Symptoms 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Depression 8a/CAT – 18-20 yr olds											X

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Child Emotional and Behavioral functioning	Child Behavior Checklist (CBCL) for 1.5-5 yr olds							X				
	Child Behavior Checklist (CBCL) for 6-18 yr olds									X[q]	X	
Child Social Cognition	Social Responsiveness Scale – Second Edition (SRS-2) School Age – Short Form									X	X	
Child Perceived Stress	PROMIS v1.0 – Psychological Stress Experiences 4a – Parent Proxy/CAT										X	
	PROMIS v1.0 – Psychological Stress Experiences 4a – Pediatric/CAT										X	
	NIH Toolbox v2.0 – Perceived Stress Scale 10-item – 18-20 yr olds											X
Concept: Child Social Role Performance and Functioning												
Peer Relationships	PROMIS v1.0 – Social Relationships – Peer Relationships 4a – Early Childhood Parent Report								X			
	PROMIS v2.0 – Peer Relationships 7a – Parent Proxy/CAT										X	
	PROMIS v2.0 – Peer Relationships 8a – Pediatric/CAT										X	
	PROMIS v2.0 – Companionship 6a – 18-20 yr olds											X
Child Social Support	NIH Toolbox v2.0 – Emotional Support – 8-17 yr olds/CAT										X	
	NIH Toolbox v2.0 – Emotional Support -18-20 yr olds/CAT											X
Bullying/Victimization	Healthy Pathways Bullying Scales										X	

[q] If the child is not yet 6 when entering the 6-10 age band, administer the CBCL 1.5-5yr instead.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Academic Ability	Early Academic Competencies – Parent Report[r]								X			
Concept: Child Well-being												
Child Global Health	PROMIS v1.0 – Global Health Scale 8a – Early Childhood Parent Report							X	X			
	PROMIS v1.0 – Global Health Scale 7+2 – Parent Proxy										X	
	PROMIS v1.0 – Global Health Scale 7+2 – Pediatric										X	
	PROMIS v1.2 – Global Health Scale – 18-20 yr olds											X
Child Curiosity	PROMIS v1.0 – Engagement – Curiosity 6a – Early Childhood Parent Report							X	X			
Child Flexibility	PROMIS v1.0 – Self-Regulation – Flexibility 5a – Early Childhood Parent Report							X	X			
Child Life Satisfaction	PROMIS v1.0 – General Life Satisfaction 5a – 18-20 yr olds/CAT											X
Child Meaning and Purpose	PROMIS v1.0 – Meaning and Purpose 8a – Pediatric/CAT										X	
	PROMIS v1.0 – Meaning and Purpose 8a – 18-20 yr olds/CAT											X
Child Self-Worth	Harter Global Self-Worth Scale – Adolescent										X	X
Child Positive Affect	PROMIS v1.0 – Positive Affect – Early Childhood Parent Report 4a/CAT							X	X			
	PROMIS v1.0 – Positive Affect – Pediatric 4a/CAT										X	
	PROMIS v1.0 – Positive Affect 4a/CAT											X

[r] Adapted from the 2016 National Survey of Children's Health conducted by the United States Census Bureau, US Department of Health and Human Services.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Concept: Partner-Related												
Verification of Current Partner Status	Partner – Pre-visit Screening Questionnaire[s]					X						
Partner Demographics	Demographics of the Partner					X						
Partner Weight and Height	Partner Weight and Height – Self Report					X						
Partner Medical History	Medical History of the Partner					X						
Partner Supplements	Supplements – Partner					X						
Partner Pregnancy Status and Intentions	Pregnancy Intentions – Partner					X						
Partner Occupation and Employment	Caregiver Occupation and Employment – Partner					X						
Partner Racial and Ethnic Identity	Racial/Ethnic Identity[c]					X						
Partner Skin Tone Assessment	Skin Tone Assessment[d]					X						
Partner Discrimination	The Everyday Discrimination Scale					X						
Partner Stressful Life Events In Childhood	Adverse Childhood Experiences					X						
Partner Positive Life Events in Childhood	Benevolent Childhood Experiences					X						
Partner Substance Use	Lifestyle – Current[f]					X						

[s] The Partner uses this form to confirm they are still the current partner of the previously pregnant person, before completing other partner collections in the 6-11 month visit.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Partner Depressive Symptoms	PROMIS v1.0 – Depression 8a/CAT					X						
Partner Anxiety	PROMIS v1.0 – Anxiety 8a/CAT					X						
Concept: Child Biospecimens												
Cord Blood				X								
Child Urine					X							
Child Stool					X							
Child Blood Spot					X		X					
Child Hair							X		X			
Child Shed Teeth[t]									X	X	X	
Concept: Maternal Biospecimens												
Maternal Saliva for DNA			X[u]		X[v]							
Maternal Whole Blood		X	X									

[t] Teeth collected will be shed deciduous (baby) teeth. There is no "age limit" for a tooth specimen, i.e., there is no upper or lower cutoff for age at which tooth was shed, nor how long it's been since it has fallen out.

[u] Collect Saliva for DNA ONLY if the pregnant participant does not provide either of the two prenatal whole blood collections.

[v] Collect Saliva for DNA ONLY if participant does not provide either of the two prenatal whole blood collections AND does not provide Saliva during the Prenatal 3rd Trimester visit.

Core Data Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr [a]	11-17 yr	18-20 yr
Maternal Blood Spot					X[w]		X[w]					
Maternal Urine		X	X			X[w]		X[w]				
Placenta				X								
Maternal Hair					X[w]							
Concept: Partner Biospecimens[x]												
Partner Saliva						X						
Partner Urine						X						
Partner Sperm						X						

[w] Collect specimens from the biological mother following end of Cycle 2 pregnancy.
[x] Collect specimens from the opposite sex partner of the biological mother following the end of Cycle 2 pregnancy.

Table 9 Specialized Data Elements for Year 2025

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation							X				
Personal Care Products	Personal Care Products - Pregnancy		X									
	Personal Care Products – Child					X						
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness Pregnancy		X									
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					
Household Water Sample	Household water sample	X										

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X				
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire- Short Form[f]					X						
Neighborhood characteristics	About My Neighborhood - Pregnancy	X										
	About My Neighborhood - Child						X		X			
Greenspace Utilization	Greenspace Utilization - Pregnancy	X										
	Greenspace Utilization - Child						X		X			

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X						
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X		X			
Family Environment	Family Environment Scale[j]	X				X	X	X	X			
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X				
Home Learning Environment	Home Learning Environment - Infancy					X		X				

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Partner Home Learning Environment	Home Learning Environment - Partner					X						
Home Learning Environment and Family Routines	Home Learning Environment - Early Childhood								X			
Family Resilience	Family Resilience[k]					X		X	X			
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X		X			
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
Child ADHD screener	SNAP-IV - Parent Report[l]									X	X	X
	Current Symptoms Scale - Self Report[m]										X	X

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

[l] Swanson, J., Nolan, W., & Pelham, W. E. (1992). The SNAP-IV rating scale. Irvine, CA: University of California at Irvine. Swanson J. M., Schuck S., Porter M. M., Carlson C., Hartman C. A., Sergeant J. A., Lakes K. (2012). Categorical and dimensional definitions and evaluations of symptoms of ADHD: History of the SNAP and the SWAN Rating Scales. International Journal of Educational and Psychological Assessment, 10(1), 51-70.

[m] From Attention-Deficit Hyperactivity Disorder: A Clinical Workbook (3rd ed.) by Russell A. Barkley and Kevin R. Murphy. Copyright 2006 by The Guilford Press.

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Temperament	Rothbart Childhood Behavior Questionnaire - Very Short Form[n]									X		
	Early Adolescent Temperament Questionnaire - Revised - Parent Report[o]									X		
	Early Adolescent Temperament Questionnaire - Revised - Self Report[o]									X	X	
	Adult Temperament Questionnaire[p]											X
	Affective Reactivity Index - Parent Report[q]									X	X	
	Affective Reactivity Index - Self Report[q]									X	X	
	Brief Sensation Seeking Scale for Children[r]									X	X	

[n] Modified from: Rothbart, M. K., Ahadi, S. A., Hershey, K. L., & Fisher, P. (2001). Investigations of temperament at 3-7 years: The Children's Behavior Questionnaire. *Child Development, 72*, 1394-1408.Putnam, S. P., & Rothbart, M. K. (2006). Development of Short and Very Short forms of the Children's Behavior Questionnaire. *Journal of Personality Assessment, 87 (1)*, 103-113.

[o] Ellis, L. K. (2002). Individual differences and adolescent psychological development. Unpublished doctoral dissertation, University of Oregon; Ellis, L. K., & Rothbart, M. K. (2001). Revision of the Early Adolescent Temperament Questionnaire. Poster presented at the 2001 Biennial Meeting of the Society for Research in Child Development, Minneapolis, Minnesota; Capaldi, D. M. & Rothbart, M. K. (1992). Development and Validation of an Early Adolescent Temperament Measure. Journal of Early Adolescence, 12,153-173.

[p] Derryberry, D. & Rothbart, M.K. (1988). Arousal, affect, and attention as components of temperament. Journal of Personality and Social Psychology, 55, 958-966; Rothbart, M.K. Ahadi, S.A. & Evans D.E. (2000). Temperament and personality: Origins and outcomes. Journal of Personality and Social Psychology, 78, 122-135

[q] Stringaris A, Goodman R, Ferdinando S, Razdan V, Muhrer E, Leibenluft E, Brotman MA. The Affective Reactivity Index: a concise irritability scale for clinical and research settings. Journal of Child Psychology and Psychiatry. 2012 Nov;53(11):1109-17.

[r] Jensen JD, Weaver AJ, Ivic R, Imboden K. Developing a brief sensation seeking scale for children: Establishing concurrent validity with video game use and rule-breaking behavior. Media Psychology. 2011 Mar 9;14(1):71-95.

Specialized Element for Year 2025	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Obesity												
Child skin folds	Child skin folds				X	X	X	X	X	X		
Child Waist circumference	Child Waist circumference								X	X	X	X
Child stool	Child stool sample					X	X	X				
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X				

Table 10 Specialized Data Elements for Year 2026

Specialized Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation						X		X			
Personal Care Products	Personal Care Products – Pregnancy		X									
	Personal Care Products - Child					X			X			
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness – Pregnancy		X									
	Natural Disasters and Preparedness - Child								X			
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Household Water Sample	Household water sample	X										
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X	X			
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire - Short Form[f]					X						
	Comprehensive Feeding Practices Questionnaire - Short Form								X			
Neighborhood characteristics	About My Neighborhood-Pregnancy	X										
	About My Neighborhood-Child						X					

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010

Specialized Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Greenspace Utilization	Greenspace Utilization-Pregnancy	X										
	Greenspace Utilization Child						X					
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X			X			
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X					
Family Environment	Family Environment Scale[j]	X				X	X	X				
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X				

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Home Learning Environment	Home Learning Environment - Infancy					X		X				
Partner Home Learning Environment	Home Learning Environment - Partner					X						
Family Resilience	Family Resilience[k]					X		X				
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X					
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
	CELF Sentence Memory								X	X		
Child ADHD screener	Preschool ADHD Rating Scale[l]								X			

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

[l] McGoey, K. E., DuPaul, G. J., Haley, E., Shelton, T. L. (2007). Parent and teacher ratings of attention-deficit/hyperactivity disorder in preschool: The ADHD rating scale-IV preschool version. Journal of Psychopathological and Behavioral Assessment, 29:269-276.

Specialized Element for Year 2026	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Temperament	Affective Reactivity Index - Parent Report[m]								X			
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Academic Ability	Reading									X	X	X
	Math								X	X	X	X
	Science										X	X
Obesity												
Weight stigma	Weight stigma										X	
Child skin folds	Child skin folds				X	X	X	X	X			
Child Waist circumference	Child Waist circumference											X
Child stool	Child stool sample					X	X	X	X	X	X	X
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X	X	X	X	

[m] Stringaris A, Goodman R, Ferdinando S, Razdan V, Muhrer E, Leibenluft E, Brotman MA. The Affective Reactivity Index: a concise irritability scale for clinical and research settings. Journal of Child Psychology and Psychiatry. 2012.

Table 11 Specialized Data Elements for Year 2027

Specialized Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation							X				
Personal Care Products	Personal Care Products Pregnancy		X									
	Personal Care Products - Child					X						
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness - Pregnancy		X									
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					
Household Water Sample	Household water sample	X										

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X				
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire- Short Form[f]					X						
Neighborhood characteristics	About My Neighborhood - Pregnancy	X										
	About My Neighborhood - Child						X					
Greenspace Utilization	Greenspace Utilization - Pregnancy	X										
	Greenspace Utilization - Child						X					

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010

Specialized Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X						
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X					
Family Environment	Family Environment Scale[j]	X				X	X	X				
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X	X			
Home Learning Environment	Home Learning Environment - Infancy					X		X				

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Partner Home Learning Environment	Home Learning Environment - Partner					X						
Family Resilience	Family Resilience[k]					X		X				
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X					
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Obesity												
Child skin folds	Child skin folds				X	X	X	X	X			
Child Waist circumference	Child Waist circumference								X	X	X	

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

Specialized Element for Year 2027	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Whole Blood	Child Whole Blood									X	X	
Child stool	Child stool sample					X	X	X				
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X				

Table 12 Specialized Data Elements for Year 2028

Specialized Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation							X				
Personal Care Products	Personal Care Products – Pregnancy		X									
	Personal Care Products - Child					X						
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness – Pregnancy		X									
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					
Household Water Sample	Household water sample	X										

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X				
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire- Short Form[f]					X						
Neighborhood characteristics	About My Neighborhood-Pregnancy	X										
	About My Neighborhood-Child						X		X			
Greenspace Utilization	Greenspace Utilization Pregnancy	X										
	Greenspace Utilization-Child						X		X			

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010.

Specialized Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X			X			
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X		X			
Family Environment	Family Environment Scale[j]	X				X	X	X	X			
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X				
Home Learning Environment	Home Learning Environment - Infancy					X		X				
Partner Home Learning Environment	Home Learning Environment - Partner					X						

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Home Learning Environment and Family Routines	Home Learning Environment - Early Childhood								X			
Family Resilience	Family Resilience[k]					X		X	X			
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X		X			
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Obesity												
Weight stigma	Weight stigma										X	
Child skin folds	Child skin folds				X	X	X	X	X			

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

Specialized Element for Year 2028	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child Waist circumference	Child Waist circumference								X	X	X	X
Child stool	Child stool sample					X	X	X				
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X				

Table 13 Specialized Data Elements for Year 2029

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation							X	X			
Personal Care Products	Personal Care Products - Pregnancy		X									
	Personal Care Products - Child					X			X			
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness - Pregnancy		X									
	Natural Disasters and Preparedness - Childhood								X			
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Household Water Sample	Household water sample	X										
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X	X			
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire- Short Form[f]					X						
	Comprehensive Feeding Practices Questionnaire Short Form								X			
Neighborhood characteristics	About My Neighborhood-Pregnancy	X										
	About My Neighborhood-Child						X					

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Greenspace Utilization	Greenspace Utilization-Pregnancy	X										
	Greenspace Utilization-Child						X					
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X						
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X					
Family Environment	Family Environment Scale[j]	X				X	X	X				
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X				

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Home Learning Environment	Home Learning Environment - Infancy					X		X				
Partner Home Learning Environment	Home Learning Environment - Partner					X						
Family Resilience	Family Resilience[k]					X		X				
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X					
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
	CELF Sentence Memory								X	X		

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child ADHD screener	Preschool ADHD Rating Scale[l]								X			
	SNAP-IV- Parent Report[m]									X	X	X
	Current Symptoms Scale - Self Report[n]										X	X

[l] McGoey, K. E., DuPaul, G. J., Haley, E., Shelton, T. L. (2007). Parent and teacher ratings of attention-deficit/hyperactivity disorder in preschool: The ADHD rating scale-IV preschool version. Journal of Psychopathological and Behavioral Assessment, 29:269-276.

[m] Swanson, J., Nolan, W., & Pelham, W. E. (1992). The SNAP-IV rating scale. Irvine, CA: University of California at Irvine. Swanson J. M., Schuck S., Porter M. M., Carlson C., Hartman C. A., Sergeant J. A., Lakes K. (2012). Categorical and dimensional definitions and evaluations of symptoms of ADHD: History of the SNAP and the SWAN Rating Scales. International Journal of Educational and Psychological Assessment, 10(1), 51-70.

[n] From Attention-Deficit Hyperactivity Disorder: A Clinical Workbook (3rd ed.) by Russell A. Barkley and Kevin R. Murphy. Copyright 2006 by The Guilford Press.

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Temperament	Rothbart Childhood Behavior Questionnaire - Very Short Form[o]									X		
	Early Adolescent Temperament Questionnaire - Revised - Parent Report[p]									X		
	Early Adolescent Temperament Questionnaire - Revised - Self Report[p]									X	X	
	Adult Temperament Questionnaire[q]											X
	Affective Reactivity Index - Parent Report[r]								X	X	X	
	Affective Reactivity Index - Self Report[r]									X	X	
	Brief Sensation Seeking Scale for Children[s]									X	X	

[o] Modified from: Rothbart, M. K., Ahadi, S. A., Hershey, K. L., & Fisher, P. (2001). Investigations of temperament at 3-7 years: The Children's Behavior Questionnaire. *Child Development, 72*, 1394-1408.Putnam, S. P., & Rothbart, M. K. (2006). Development of Short and Very Short forms of the Children's Behavior Questionnaire. *Journal of Personality Assessment, 87 (1)*, 103-113.

[p] Ellis, L. K. (2002). Individual differences and adolescent psychological development. Unpublished doctoral dissertation, University of Oregon; Ellis, L. K., & Rothbart, M. K. (2001). Revision of the Early Adolescent Temperament Questionnaire. Poster presented at the 2001 Biennial Meeting of the Society for Research in Child Development, Minneapolis, Minnesota; Capaldi, D. M. & Rothbart, M. K. (1992). Development and Validation of an Early Adolescent Temperament Measure. Journal of Early Adolescence, 12,153-173.

[q] Derryberry, D. & Rothbart, M.K. (1988). Arousal, affect, and attention as components of temperament. Journal of Personality and Social Psychology, 55, 958-966; Rothbart, M.K. Ahadi, S.A. & Evans D.E. (2000). Temperament and personality: Origins and outcomes. Journal of Personality and Social Psychology, 78, 122-135

[r] Stringaris A, Goodman R, Ferdinando S, Razdan V, Muhrer E, Leibenluft E, Brotman MA. The Affective Reactivity Index: a concise irritability scale for clinical and research settings. Journal of Child Psychology and Psychiatry. 2012 Nov;53(11):1109-17.

[s] Jensen JD, Weaver AJ, Ivic R, Imboden K. Developing a brief sensation seeking scale for children: Establishing concurrent validity with video game use and rule-breaking behavior. Media Psychology. 2011 Mar 9;14(1):71-95.

Specialized Element for Year 2029	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Academic Ability	Reading									X	X	
	Math								X	X	X	
	Science										X	
Obesity												
Child skin folds	Child skin folds				X	X	X	X	X	X		
Child Waist circumference	Child Waist circumference									X	X	X
Child stool	Child stool sample					X	X	X		X	X	X
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X	X	X	X	

Table 14 Specialized Data Elements for Year 2030

Specialized Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Exposure Area												
Chemical/Physical												
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Food Contaminants	Child Food Source and Preparation							X				
Personal Care Products	Personal Care Products - Pregnancy		X									
	Personal Care Products - Child					X						
Climate	Air Quality Awareness and Protective Behaviors		X									
	Natural Disasters and Preparedness - Pregnancy		X									
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Child Urine	Child Urine sample						X					
Household Water Sample	Household water sample	X										

[a] Child self report begins at age 8 years, such that any child self report measure collected in the 6-10 year age band is only applicable to those children who are 8-10 years of age. Cohort Study Sites collecting data on 6-7-year-olds in this age band do not administer the child self report instruments.

[b] National Cancer Institute. Dietary Screener Questionnaire (DSQ) in the NHANES 2009–10: DSQ. Available from: http://epi.grants.cancer.gov/nhanes/dietscreen/questionnaires.html.

Specialized Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Lifestyle												
Maternal Eating Behavior	Meal Patterning[c]	X										
Child Appetite	Baby Eating Behavior Questionnaire[d]					X	X					
	Child Eating Behavior Questionnaire[e]							X				
Child Diet	Dietary Screener Questionnaire - Parent Report[b]							X				
Child Feeding	Infant Feeding Style Questionnaire- Short Form[f]					X						
Neighborhood characteristics	About My Neighborhood - Pregnancy	X										
	About My Neighborhood - Child						X					
Greenspace Utilization	Greenspace Utilization - Pregnancy	X										
	Greenspace Utilization - Child						X					

[c] Alfonsson S, Sewall A, Lidholm H, Hursti T. The Meal Pattern Questionnaire: A psychometric evaluation using the Eating Disorder Examination. Eat Behav. 2016 Apr;21:7-10. doi: 10.1016/j.eatbeh.2015.12.002. Epub 2015 Dec 18. PMID: 26722817.

[d] Modified from Llewellyn CH, van Jaarsveld CH, Johnson L, Carnell S, Wardle J. Development and factor structure of the Baby Eating Behaviour Questionnaire in the Gemini birth cohort. Appetite. 2011 Oct;57(2):388-96.

[e] Modified from Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. The Journal of Child Psychology and Psychiatry and Allied Disciplines. 2001 Oct;42(7):963-70. Carnell S, Wardle J. Measuring behavioural susceptibility to obesity: validation of the child eating behaviour questionnaire. Appetite. 2007 Jan 1;48(1):104-13.

[f] Modified from the Infant Feeding Styles Questionnaire: Thompson AL, Mendez MA, Borja JB, Adair LS, Zimmer CR, Bentley ME. Development and validation of the Infant Feeding Style Questionnaire. Appetite. 2009 Oct;53(2):210-21. doi:10.1016/j.appet.2009.06.010.

Specialized Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Maternal Breastmilk	Maternal Breastmilk Collection Screening				X							
	Maternal Breastmilk sample				X							
Psychosocial												
Birthing Experience	Birthing Experience				X							
Caregiver Birth Traumatic Experiences	Child Birth Trauma Scale[g]					X						
Partner Traumatic Birth Experiences	Child Birth Trauma Scale - Partner[g]					X						
Caregiver Discrimination During Medical Care	Discrimination in Medical Settings[h]	X				X						
Partner Discrimination During Medical Care	Discrimination in Medical Settings[h]					X						
Climate Change Anxiety	Climate Change Anxiety Scale[i]		X				X					
Family Environment	Family Environment Scale[j]	X				X	X	X				
Parent-Child Interaction	Early Regulation in Context Assessment (ERICA)							X	X			
Home Learning Environment	Home Learning Environment - Infancy					X		X				
Partner Home Learning Environment	Home Learning Environment - Partner					X						

[g] City Birth Trauma Scale © Ayers, Wright & Thornton 2018. Frontiers in Psychiatry 9:409.

[h] Peek ME, Nunez-Smith M, Drum M, Lewis TT. Adapting the everyday discrimination scale to medical settings: reliability and validity testing in a sample of African American patients. Ethn Dis. 2011 Autumn;21(4):502-9.

[i] Clayton S, Karazsia BT. Development and validation of a measure of climate change anxiety. Journal of environmental psychology. 2020 Jun 1;69:101434.

[j] FES Instrument © 1974, 2002 Rudolph H. Moos. All rights reserved in all media. Published by Mind Garden, Inc., www.mindgarden.com

Specialized Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Family Resilience	Family resilience[k]					X		X				
Caregiver Religiosity/Spirituality	Religiosity and Spirituality	X					X					
Outcome Area												
Neurodevelopment												
Child General Cognition	NIH Baby Toolbox Executive Function						X	X				
	NIH Baby Toolbox Looking While Listening						X					
Child Language Development	NIH Baby Toolbox MacArthur Bates CDI - CAT Production						X					
	NIH Baby Toolbox Picture Vocabulary							X				
Motor development	NIH Baby Toolbox Get up and Go						X					
	NIH Baby Toolbox Motor Reach to Eat						X					
Obesity												
Weight stigma	Weight stigma										X	
Child skin folds	Child skin folds				X	X	X	X	X			
Child Waist circumference	Child Waist circumference								X			

[k] National Survey of Children's Health. Center for Child and Adolescent Health Measurement Initiative. 2023. Bethell CD, Gombojav N, Whitaker RC. Family Resilience And Connection Promote Flourishing Among US Children, Even Amid Adversity. Health Aff (Millwood). 2019 May;38(5):729-737. doi: 10.1377/hlthaff.2018.05425. PMID: 31059374.

Specialized Element for Year 2030	Measure	Prenatal Early Pregnancy	Prenatal 3rd trimester	Perinatal	0-5 mo	6-11 mo	12-23 mo	24-35 mo	3-5 yr	6-10 yr[a]	11-17 yr	18-20 yr
Child stool	Child stool sample					X	X	X	X			
Maternal stool	Maternal stool sample		X									
Upper and Lower Airways												
Asthma Control	Asthma Control and Medication Adherence - Parent Report									X		
	Asthma Control and Medication Adherence - Self Report										X	
Nasal swab	Nasal swab sample				X		X	X				

5. Data Management

5.1 Data Submission to the ECHO Data Analysis Center (DAC)

The ECHO Program expects Cohort Study Sites to transfer all newly collected data elements included in the ECHO Cohort Data and Biospecimen Collection Protocol to the ECHO DAC in a manner consistent with the ECHO Data Sharing and Use Policy.

Cohort Study Sites will submit all data to the ECHO DAC through ECHOPortal.org, the main hub for data-centered ECHO Cohort activities. The DAC will associate users of ECHOPortal with their appropriate awarded Cohort Study Site(s) and set default access for the users to view data. In addition, each user will be assigned one or more authorization groups or roles (e.g., data manager, study coordinator) that govern the access granted for various utilities and modules on ECHOPortal.

The ECHO DAC will provide an electronic data capture system for the collection of all data elements included in the ECHO Cohort Data and Biospecimen Collection Protocol through “REDCap Central” on ECHOPortal. REDCap (Research Electronic Data Capture) is a metadata-driven electronic data capture software for web-based application developed by Vanderbilt University. Data entered by the Cohort Study Site into this system will be captured in real-time in a central secure database at the ECHO DAC.

Cohort Study Sites may collect some data elements by third-party applications such as NIH Toolbox, in which case, the Cohort Study Site users with approved uploading privileges will use the “Data Upload” utility within ECHOPortal to transfer such data.

5.2 ECHOPortal and Data Security

ECHOPortal is a web portal to a High-Performance Computing infrastructure developed and maintained by the ECHO DAC to receive, host, analyze, and control access to data. ECHOPortal provides access to numerous tools, including a metadata catalog of data collected by each cohort site and available on the ECHO Cohort Data Platform, a participant registration system, data capture and processing systems, a biospecimen tracking system, and tools for the analysis of ECHO Cohort data.

ECHO Cohort Study Sites will collect data under the ECHO Cohort Data and Biospecimen Collection Protocol and enter or transfer these data through ECHOPortal. The system and tools available through ECHOPortal that ECHO Cohort Study Sites will use to collect data under this protocol comply with the Federal Information Security Management Act (FISMA) low-level information security requirements for federal information security systems.

Data will be processed, stored, and analyzed at the ECHO DAC under FISMA moderate security controls. FISMA moderate security requires that the ECHO DAC use extensive security procedures and controls as well as regular vulnerability scans of the computers and related systems. ECHO DAC annual security certification and accreditation reviews are performed by

the DAC's Global Technology Solutions (GTS) – Office of the Chief Information Security Officer to ensure continued compliance with the Authority to Operate. Additionally, GTS has received an Authority to Operate under the National Institute of Standards and Technology (NIST) SP 800-53r4 for Federal Information Processing Standards (FIPS) LOW and FIPS MOD classifications assessed by an accredited FedRAMP Third Party Assessment Organizations (3PAO). The following required security measures are implemented to decrease the risk of a breach or data leak:

- Multiple network firewalls and an Intrusion Prevention/Detection system protect all network communication in and out of the moderate network. The network implements a "default deny" rule that drops all traffic not explicitly permitted. The ECHO DAC monitors network traffic for malicious behavior and other specific attack signatures.
- Access to the moderate network is controlled through a virtual private portal that requires multi-factor authentication using a software token and a PIN. All connections to the moderate network require an encrypted virtual private network connection that securely encrypts all network traffic and data in transit.
- Telnet, File Transfer Protocol, Simple Mail Transfer Protocol, Network Basic Input/Output System, and other clear-text (not encrypted) transmissions to the Internet are not permitted.
- All outbound emails from a FISMA moderate security network are limited to 50KB in size and emails cannot include attachments.
- Users cannot take information from the environment and paste it into an external system or application.

Within ECHOPortal, data are encrypted when not being used. Data transmissions from the ECHO Cohort Study Sites will be encrypted and captured in a secure data lake that is also encrypted within the ECHOPortal environment. The ECHO Data Lake is a write-once, then read-only data storage, allowing the data to be tracked exactly as submitted, including information regarding when the data was submitted and by whom.

5.3 Data Quality Assurance

The ECHO DAC, the CC, the Cohort Study Sites, Laboratory Core, and Measurement Core, and other involved ECHO components will follow written standard operating procedures documented in the ECHO MOP to ensure that data are collected in compliance with the ECHO Cohort Data and Biospecimen Collection Protocol. The ECHO Program will also employ quality assurance (QA) and quality control (QC) procedures documented in a Data Quality Plan to assure that the data and reported results are credible and accurate. Quality assurance activities refer to steps taken prior to the collection of data, and QC activities refer to steps taken during and following data collection to ensure unified data collection standards across ECHO Cohort Study Sites.

The QA/QC procedures for data collected under this protocol include the following:

- Development of the ECHO Cohort Data and Biospecimen Collection Protocol, data collection forms, data dictionaries, and the MOP.
- Development of an electronic data capture system available to Cohort Study Sites that includes automated edit checks to identify potential errors at the point of entry, participant scheduling, and reporting needs.
- Rigorous testing of the data capture system capabilities and functioning, prior to release, and establishment of change control processes, issue tracking, and load testing.
- Training and certification of Cohort Study Site staff in measurement and data collection procedures.
- Regular communication between the ECHO CC/DAC and Cohort Study Site staff to review progress and address questions and concerns.

The QC metrics for data collected under this protocol include the following:

- Numbers of children and pregnant persons enrolled and completeness of follow-up by Cohort Study Site and overall.
- Number and percent of variable values that are missing, outside of accepted ranges, or inconsistent with other data provided.
- Comprehensive quality (e.g., illogical values, unexpected distributional properties) and completeness (at item and form level) assessment.

5.4 Data Access

ECHO Cohort data may be accessed within a secure FISMA moderate virtual computing enclave that includes two-factor authentication (i.e., identity must be proven in two different ways before a user can gain access). ECHO analysts may access dates that are individually unique and pose a theoretical risk of participant re-identification, but only DAC staff (e.g., data analysts, system administrators) may access individuals' geographic locations.

The policies and procedures for obtaining access to ECHO Cohort data to conduct analyses for an ECHO Cohort Steering Committee-approved analysis proposal are described in the ECHO Data Sharing and Use Policy, the ECHO Publications Policy, and the ECHO MOP.

The ECHO DAC may revoke any and all access for users if they are found to violate the policies, principles, and procedures of the NIH and the ECHO Data Sharing and Use Policy. This policy ensures that security controls meet government standards and that security risks have appropriate mitigation strategies. Security controls in place include access control, physical and environmental protection, personnel security, accountability and audit, and system/information integrity.

5.5 Biospecimen Tracking

The ECHOPortal Biospecimen Tracking system (Bio-Track) will manage biospecimens and their associated metadata. Within this system, Cohort Study Site staff enter information about the collection, processing, and storage steps taken with each biospecimen at their local facilities and link the biospecimen to a participant and to their location in local storage facilities. In addition, staff enter relevant information collected about the participant (e.g., time since the participant last ate). Authorized users can perform specimen container and sample management, prepare shipment manifests, and finalize shipments using Bio-Track. Shipment information, containing only de-identified specimens, container information, and shipping information is securely transmitted to the central repository's Laboratory Information Management System (LIMS) using Bio-Track.

5.6 Record Retention

Data transmitted or entered via ECHOPortal under the ECHO Cohort Data and Biospecimen Collection Protocol will be maintained at the ECHO DAC for the duration of the ECHO DAC grant. Throughout the cycle, the ECHO DAC will transfer de-identified data to an NIH-approved repository consistent with the ECHO Data Sharing and Use Policy.

6. Analysis Proposals

The objective of this protocol is to standardize data and biospecimen collection and sharing across the ECHO Program.

Separate, research question-specific analysis proposals will describe future analyses of data collected under the ECHO Cohort Protocol (i.e., analyses of the ECHO Cohort data). ECHO Investigators will develop these analysis proposals in close collaboration with the ECHO DAC, and the ECHO Publications Committee will review and approve them (see the ECHO Publications Policy for more information).

7. Study Responsibilities

7.1 ECHO Coordinating Center Responsibilities

The ECHO CC will provide the overall coordination and support of activities and communications accruing to the conduct of the ECHO Cohort Data and Biospecimen Collection Protocol. The ECHO CC is responsible for maintaining the current version of the ECHO Cohort Data and Biospecimen Collection Protocol and will make any protocol revisions available to the ECHO Cohort Study Sites following ECHO single Institutional Review Board (sIRB) approvals.

In addition, the ECHO CC will:

- Serve as the main point of contact between the ECHO single Institutional Review Board (sIRB) and the Cohort Study Sites, and facilitate event reporting from the cohort sites and the ECHO DAC to the ECHO sIRB.
- Develop and maintain the ECHO MOP.
- Facilitate the ECHOPortal access request process.
- Collaborate with the ECHO DAC, the ECHO Measurement Core, experienced ECHO Cohort Study Sites, and the ECHO Laboratory Core to provide protocol training.
- Review all ECHO Cohort Study Site-specific informed consent documents to ensure compliance with the ECHO Cohort Data and Biospecimen Collection Protocol and related policies.
- Collaborate with the ECHO DAC and the ECHO Measurement Core regarding data quality assurance by ensuring the administration of outcome tools and acquisition of biospecimens is appropriate and consistent across all research sites.

7.2 ECHO Data Analysis Center Responsibilities

The ECHO DAC is responsible for developing and maintaining an environment to host, analyze, and control access to ECHO Cohort data. The environment is accessible through a web portal (ECHOPortal.org) that provides capabilities to receive data from ECHO Cohort Study Sites using secure transmission protocols, capture newly collected individual-level data using web-based data collection tools, and query, visualize, and analyze harmonized ECHO Cohort data. The DAC is also responsible for tracking biospecimens and their associated metadata in the ECHOPortal Bio-Track. The DAC will promote data on the ECHO Cohort Data Platform that are of high quality (valid, reliable, and complete) following activities, processes and procedures documented in the ECHO Data Quality Plan.

7.3 ECHO Measurement Core Responsibilities

The ECHO Measurement Core is responsible for efforts to enable, curate, and, where needed, enhance existing or develop new measures of exposures, covariates, and outcomes obtained through self- or proxy-report, observation, performance-based testing, or personal sensor devices. The ECHO Measurement Core will conduct quality checks on measurement data and provide support for data harmonization and analyses. The ECHO Measurement Core will also provide translation into Spanish and other languages as needed for participant-facing instruments.

7.4 ECHO Cohort Study Site Investigator Responsibilities

It is the responsibility of the ECHO Cohort Study Site Investigator or designee with oversight for any participating location to:

- Provide all study personnel under their supervision with copies of the ECHO Cohort Data and Biospecimen Collection Protocol and access to all information provided by the ECHO CC, and ensure that they are fully informed about the conduct of the processes within the protocol.
- Ensure that a properly constituted IRB has reviewed and approved the ECHO Cohort Data and Biospecimens Collection Protocol through a reliance agreement to the sIRB prior to the initiation of any protocol-related procedures.
- Ensure that prior to the initiation of any protocol-related procedures, research staff have engaged each participant or legal guardian in an informed consent process that involved full and adequate information about the protocol, and that the participant or legal guardian has signed and dated an informed consent (or assent, as appropriate) form, as required by the IRB of record.
- Retain source documents including, but not limited to, the signed and dated informed consent document, study-related progress notes, completed interviews, and questionnaires, study or participant-specific correspondences, records of files created and submitted to the ECHO DAC, according to the IRB of record/institution and NIH policies regarding record retention.
- Collect and share data with the ECHO Cohort Data Platform and collect and ship biospecimens to the ECHO-designated biorepository under the ECHO Cohort Data and Biospecimen Collection Protocol.

7.5 ECHO Laboratory Core Responsibilities

ECHO Laboratory Core provides leadership and infrastructure for all activities related to ECHO Cohort biospecimens and data generated from biospecimen assays. The main roles of the ECHO Laboratory Core are to facilitate the collection and processing of biospecimens; manage the ECHO Cohort Biorepository; perform or facilitate a wide range of biospecimen assays to support ECHO Cohort analyses; and coordinate biospecimen information and assay results.

8. Ethical Considerations and Human Subjects Protection

8.1 Institutional Review Board/Ethics Committee Review

A properly constituted IRB of record, formally designated to review and monitor research involving human subjects, will be accountable for compliance with regulatory requirements for the ECHO Cohort Data and Biospecimen Collection Protocol at participating Cohort Study Sites. Participating Cohort Study Sites will rely on the sIRB to satisfy the regulatory requirements relevant to the IRB review. The participating Cohort Study Sites will maintain essential required

documentation of IRB reviews. A reliance agreement will be required between the ECHO sIRB and participating Cohort Study Sites IRBs.

The sIRB will review this protocol and all informed consent/assent forms, HIPAA Authorization forms, recruitment materials, and other relevant information prior to the initiation of any ECHO Cohort Data and Biospecimen Collection Protocol-related procedures or activities. The sIRB will also review any amendments to this protocol prior to their implementation.

8.2 Informed Consent

The ECHO MOP provides detailed information regarding the ECHO Cohort Data and Biospecimen Collection Protocol informed consent process.

ECHO Cohort Study Site Investigators (or their designated study personnel) will obtain written informed consent from ECHO Cohort Data and Biospecimen Collection Protocol participants who are at least 18 years of age (or according to any state-specific regulations), or from a parent/legal guardian, along with child assent for those participants who are younger than 18 years of age (or according to any state-specific regulations). For the ECHO Cohort Data and Biospecimen Collection Protocol, the recommended lower bound for child assent is six years of age. Each participating Cohort Study Site must check local regulations and guidance to determine its lower bound. Caregiver participation may commence on the basis of caregiver consent and parental permission of the associated child, but may occur independent of assent by a child who is of the age and capacity to assent. Caregiver and parent reports are valid assessments of child behavior, health, and development that yield valuable information about child health over time, even in the absence of direct child assessments. Parental permission and child assent no longer apply when minor participants reach adulthood or legal age of majority. Such participants must reconsent to participate as adults. Electronic consent (e-consent) is also acceptable. ECHO Cohort Study Site Investigators should inform potential participants about the nature and purpose of the protocol, participation and withdrawal conditions, and the risks and benefits of participation before initiating any ECHO Cohort Data and Biospecimen Collection Protocol-related procedures or activities. The informed consent will specify that participants may choose not to participate in a specific component, biospecimen collection, questionnaire, individual question, or genetic testing at any time, and that they may stop participation in the study at any time.

ECHO Cohort Study Site Investigators will record written informed consent/assent on the properly constituted IRB-approved consent/assent/HIPAA form(s). The informed consent process and form will conform to US regulations (45 CFR § 46, the “Common Rule”) and to institutional requirements for informed consent. If applicable, the informed consent form will be provided in translation for non-English-speaking participants. ECHO Cohort Study Site Investigators will provide a printed or electronic copy of the final signed consent/document to the participant (or the parent/legal guardian).

The ECHO Program will collect brief demographic information (e.g., age, race, ethnicity, and sex) about pregnant persons and children who are eligible for the ECHO Cohort Data and

Biospecimens Collection Protocol but who do not consent. This approach will allow the ECHO Program to better understand trends and consider any biases in the resulting data.

8.3 Confidentiality of Participants

The ECHO DAC will maintain the confidentiality of human subject data obtained from participants throughout the ECHO DAC funding period. The ECHO DAC will transfer data to an NIH-approved repository at defined intervals consistent with the ECHO Data Sharing and Use Policy.

The data files submitted by ECHO Cohort Study Sites may include personal identifiers, including Protected Health Information (PHI), such as date of birth, visit dates, and residential address. The ECHOPortal environment provides an infrastructure to share data and information while respecting and protecting ECHO participant confidentiality. The alignment of different ECHO roles (e.g., cohort data manager, ECHO DAC analyst) with access to different ECHOPortal functions (e.g., data submission, data analysis) will help to safeguard participant privacy and data confidentiality. Some functions will require having a combination of the correct role and fulfillment of additional requirements, including ECHO Cohort Steering Committee approval of an analysis proposal and IRB approval for an ECHO Investigator leading an analysis proposal.

Access to certain types of data on the ECHO Cohort Data Platform within the secure FISMA moderate virtual computing enclave may be limited as specified by the Data Sharing and Use Policy if deemed appropriate to safeguard against risk of participant identification. Additional stipulations regarding data access may apply for special populations, including tribal nations.

ECHO has obtained a Federal Certificate of Confidentiality for the ECHO Cohort Data and Biospecimen Collection Protocol. The certificate will help protect against disclosures of study-related information by Federal, State, or local civil, criminal, administrative, legislative, or other proceedings, although it will not guarantee that data cannot be released. Participants will be informed about the certificate during the informed consent process.

8.4 Authorization for Use and Disclosure of Protected Health Information (HIPAA)

ECHO Cohort Investigators will ensure that the use and disclosure of PHI obtained under this protocol complies with the Health Insurance Portability and Accountability (HIPAA) Privacy Rule.

8.5 Risks to ECHO Cohort Data Collection Protocol Participants

The ECHO Cohort Data and Biospecimen Collection Protocol defines the collection of observational data. It does not include interventional procedures (e.g., the use of drugs, medical devices or procedures, vaccines, or changes of the participants' health practices). The measures and assessments in this protocol conform to the definition of minimal risk in the federal regulations (45 CFR § 46.102[j]; *the probability and magnitude of harm or discomfort anticipated in the research are not greater in and of themselves than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests*). A properly constituted IRB of record will review the measures and

assessments in this protocol under 45 CFR § 46.404, where no greater than minimal risk is identified, and provisions are made for obtaining assent from children and the permission of their parents or guardians.

8.6 Reimbursement to ECHO Cohort Data Collection Protocol Participants

ECHO Cohort Study Sites may reimburse participants for their time to participate in the ECHO Cohort Data and Biospecimen Collection Protocol. The degree of participant reimbursement will be at the discretion of each ECHO Cohort Study Site (and commensurate with participant burden).

9. Regulatory Reporting

Refer to the ECHO MOP for detailed regulatory reporting instructions.

9.1 Unanticipated Problems Involving Risks to Subjects or Others

The US Department of Health and Human Services, Office of Human Research Protections defines research-related unanticipated problems (UPs) as:

- Events that are unanticipated and related (or possibly related) to research participation; or
- Any event that, in the opinion of the investigator, puts the participant at greater risk than was previously known or recognized.

ECHO Cohort Study Site Investigators (or their designated study personnel) are responsible for reporting research-related UPs according to sIRB processes. In addition, ECHO Cohort Study Site Investigators or designated study personnel should report all research-related UPs to the ECHO CC within one business day of discovery. Cohort Study Site Investigators (or their designated study personnel) will report unanticipated problems reported to the ECHO CC, who will then track and summarize problem reports in the sIRB continuing reviews.

9.2 Breach Notifications

ECHO Cohort Study Site Investigators (or their designated study personnel) are responsible for immediately notifying the ECHO DAC and the ECHO CC of any breach in privacy and security (e.g., compromise of user account, significant virus/malware infection, or accidental inclusion of direct identifiers on records uploaded to ECHOPortal) as it may affect confidentiality, integrity, or availability of ECHOPortal data and resources. ECHO Cohort Study Site Investigators (or their designated study personnel) must also report any privacy breach according to their IRB of record/institution processes and their appropriate institutional authorities. The ECHO CC will notify the NIH ECHO Program Office within three business days of receiving the final IRB decision and may also notify the ECHO Cohort Steering Committee.

The ECHO DAC is responsible for notifying the ECHO CC, within one business day, of any breaches of privacy and security in their databases (i.e., breaches of data housed in ECHOPortal). The ECHO CC will report the breach to the ECHO sIRB and affected ECHO

Cohort Study Site Investigators within one business day of breach notification. The ECHO CC will notify the NIH ECHO Program Office within three business days of receiving the breach report and may also notify the governing ECHO Committee (e.g., the Steering Committee or Operations Committee). The ECHO CC will disseminate sIRB decisions to the ECHO DAC and affected ECHO Cohort Study Site Investigators for required actions and institutional reporting.

9.3 Reports of Non-compliance

The ECHO Program defines non-compliance as the failure to follow the determinations or requirements of the sIRB-approved protocol. ECHO Cohort Study Site Investigators (or their designated study personnel) are responsible for reporting protocol non-compliance to the ECHO CC within one business day of discovery and to the sIRB according to sIRB policies.

10. Appendices

10.1 ECHO Policies

All current ECHO Policies are located in the ECHO Program Manual under “Program Information and Governance”, “Policies.” The policies cited in the ECHO Cohort Data and Biospecimen Collection Protocol include the following:

- ECHO Publications Policy
- ECHO Biospecimen Policy
- ECHO Data Sharing and Use Policy

10.2 ECHO Cohort Data Collection Protocol Version 3.1 to Version 3.2: Summary of Changes

Sections Affected	Summary of Change	Reason for Change
Entire Protocol (global in all instances)	Changed references of Version 3.1 to Version 3.2 and updated the version dating. Added Version 3.1 to the Protocol Version and Amendment Tracking table. Added the Summary of Changes from Version 3.1 to Version 3.2 as Section (10.2). Updated Table of Contents Updated List of Tables to include Specialized Measures (Exposures and Outcomes) by Calendar year 2025-2030 Updated ECHO Cohort Data and Biospecimen Collection Protocol Glossary to include Specialized Data Element	Administrative changes required due to versioning from Version 3.1 to Version 3.2.
Protocol Synopsis **Exclusion Criteria**	Amended Exclusion Criteria as follows: **FROM**: No further exclusion criteria apply. Participants within Cohort Study Sites of ECHO Cohort Awards who provide informed consent/assent, as appropriate, are eligible for inclusion in the ECHO Cohort Data and Biospecimen Collection Protocol. **TO:** A surrogate pregnancy is excluded from enrollment.	Added text to clarify ECHO Participants, study exclusion
2.2 Development of the ECHO Cohort Data and Biospecimen Collection Protocol		
Brief Timeline of ECHO Cohort Data and Biospecimen Collection Protocol Development	Amended text to indicate “Summary of Change Table” includes a list of changed made between the prior version and the current version	Clarification

Sections Affected	Summary of Change	Reason for Change
2.3 Development of the ECHO Streamlined Protocol		
Section Title	**Amended section title** **FROM:** Development of the ECHO Cohort Data and Biospecimen Collection Protocol **TO:** Development of the ECHO Streamlined Protocol	Clarification
Body of text	**Added text to describe development of Specialized Measures /Protocol**	Inclusion of Specialized Measures
3.2 Exclusion Criteria		
Entire section	**Amended exclusion criteria** **FROM**: No further exclusion criteria apply. Participants within Cohort Study Sites of ECHO Cohort Awards who provide informed consent/assent, as appropriate, are eligible for inclusion in the ECHO Cohort Data and Biospecimen Collection Protocol. **TO:** A surrogate pregnancy is excluded from enrollment	Added text to clarify ECHO Participants, study exclusion
3.6 ECHO Cohort Data and Biospecimen Collection Protocol Design: Data Elements and Biospecimens		
Body of text	Added paragraph to describe Specialized Data Elements and Specimens	Inclusion of Specialized Measures

Sections Affected	Summary of Change	Reason for Change
4. ECHO Cohort Data and Biospecimen Collection Protocol Procedures		
Body of text	Added guidance re site selection of specialized measures	Inclusion of Specialized Measures
4.1 Frequency of Data Collection		
Body of text	Added guidance re site selection and collection of specialized measures (exposures and outcomes) **Amended Text** **FROM:** The ECHO Program expects that each Cohort Study Site will collect core data element measures and biospecimens on their cohort participants twice in the prenatal period (**before 20 weeks' gestation** and in the 3rd trimester), in the perinatal period, and at 0 to 5 months, 6 to 11 months, 12 to 23 months, and 24 to 35 months. **TO:** The ECHO Program expects that each Cohort Study Site will collect core data element measures and biospecimens on their cohort participants twice in the prenatal period (~~**before 20 weeks' gestation**~~ **early pregnancy** and in the 3rd trimester), in the perinatal period, and at 0 to 5 months, 6 to 11 months, 12 to 23 months, and 24 to 35 months.	Inclusion of Specialized Measures Align text with Data Element table age bands

Sections Affected	Summary of Change	Reason for Change
4.2 Core Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course		
Section Title **Body of text**	**Amended section title** **FROM**: **Core** Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course **TO:** ~~Core~~ Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course Referenced specialized measures Tables 9- 13	Inclusion of Specialized Measures
4.3 Specialized Data Elements and Biospecimens for New Data Collection Across the Child Developmental Life Course		
NEW section	Added text to include Specialized Elements	Inclusion of Specialized Measures
Table 2: Data Elements for Year 2024	Moved Maternal Urine collection from the 0-5 mo age band to the 6-11 mo age band	Corrected error in v3.1
Table 3, Data Elements for Year 2025 and Table 6, Data Elements for Year 2028	Added footnote to begin spirometry data collection at age 7	Per spirometry subgroup recommendation to obtain spirometry data collection starting at age 7 due to level of difficulty in younger children
Tables 2-8	Converted table endnotes to footnotes Relabeled notes **FROM**: α, β, a, ..., X **TO:** a, b, c, ..., X	Improve clarity and readability

Sections Affected	Summary of Change	Reason for Change
NEW section: **Tables 9-14**	Added Tables 9-14 to include Specialized Elements by Exposure Area and Outcome Areas collected within calendar years 2025 - 2030 Added footnotes indicating measure developers and /or copyright information	Inclusion of Specialized Measures